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Dishwasher

Abstract

A dishwasher prevents an airflow guide from clattering effectively without an additional member and supports the airflow guide reliably. A lower end surface of a fastening nut is exposed to a wash space. The dishwasher prevents wash water from remaining between the fastening nut and a tub, and prevents the tub and a dry air supply hole from corroding.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to and the benefit of Korean Patent Application Nos. 10-2021-0159662, filed on Nov. 18, 2021, and 10-2021-0159664, filed on Nov. 18, 2021, and 10-2022-0028084, filed on Mar. 4, 2022, and 10-2022-0028086, filed on Mar. 4, 2022, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present application relates to a dishwasher, and in particular, a dishwasher that can prevent

an airflow guide from clattering effectively without an additional member and support the airflow guide reliably, allow a lower end surface of a fastening nut to be exposed to a wash space, prevent wash water from remaining between the fastening nut and a tub, and prevent the tub and a dry air supply hole from corroding.

BACKGROUND

(3) Dishwashers spray wash water such as water to a wash target such as cooking vessels, cooking tools and the like accommodated in them to wash the wash target. At this time, wash water used for washing a wash target can include detergent.

(4) In some cases, dishwashers are comprised of a tub forming a wash space, a storage part accommodating wash targets in the tub, a spray arm spraying wash water to the storage part, and a sump storing water and supplying wash water to the spray arm.

(5) Dishwashers help to reduce time and efforts taken to clean wash targets such as cooking vessels and the like after meals, thereby ensuring improvement in user convenience.

(6) In some cases, dishwashers perform a washing process of washing wash targets, a rinsing process of rinsing the wash targets, and a drying process of drying the wash targets after the washing and rinsing processes.

(7) In recent years, the drying stage of dishwashers involves supplying high-temperature dry air into the tub to reduce a drying period and promote the effect of sterilizing wash targets.

(8) As a related art, a dishwasher provided with a hot air supply device that generates and supplies high-temperature dry air after the washing and rinsing stages is disclosed in DE Patent Publication No. 102015212869 (document 001).

(9) In the dishwasher according to document 001, a dry air spray part for spraying dry air generated through the hot air supply device, which is disposed under of a tub, into the tub is disposed in the tub.

(10) The dry air spray part is coupled to a hot air supply tube that extends in a way that penetrates the tub.

(11) At this time, the dry air spray part can be coupled to the hot air supply tube simply by being press-fitted to the hot air supply tube, thereby ensuring a simple assembly or fastening.

(12) However, according to document 001, without a very tight manufacturing tolerance for the dry air spray part and the hot air supply tube, clattering can occur due to a relative displacement or a relative movement that is caused by an up-down gap.

(13) Additionally, according to document 001, without rigidity of a predetermined level or above in a connecting portion between the dry air spray part and the hot air supply tube, the dry air spray part can easily escape upward and downward from its right position as quite a strong impact is applied.

(14) Further, according to document 001, without rigidity of a predetermined level or above in the connecting portion between the dry air spray part and the hot air supply tube, the connecting portion between the dry air spray part and the hot air supply tube can be readily broken as quite a strong impact is applied.

(15) Further, the dry air spray part according to document 001 extends vertically through an open hole that is formed in a way that penetrates the lower surface of the tub.

(16) Accordingly, a connecting portion between the open hole and the dry air spray part is exposed to high-temperature wash water, and when wash water remains in the connecting portion after the washing stage and the rinsing stage, the open hole and the lower surface side of the tub are highly likely to corrode.

(17) Further, it is highly likely that wash water and food included in the wash water remaining in the connection portion between the open hole and the dry air spray part can cause the reproduction of germs and generate a bad smell.

(18) Further, since the dry air spray part according to document 001 is not fixed to the hot air supply tube in a rotation direction, the dry air spray part is likely to collide with the edge of the tub

depending on the operation state of the dishwasher.
(19) Accordingly, the dry air spray part can be broken or generate noise due to its collision with the edge of the tub.
(20) Further, the dry air spray part according to document 001 has an outer shape corresponding to the shape of the edge of the tub. Accordingly, the outer shape of the dry air spray part can be big.
(21) As the dry air spray part having a big outer shape is disposed in the wash space, an available wash space of the tub can be reduced substantially, and interference with other components including a lower spray arm and a lower rack, disposed in the tub, can occur. Further, since the dry air spray part needs to be disposed closet to the tub, food and the like is highly likely to be fixed between the dry air spray part and the inner surface of the tub.
PRIOR ART DOCUMENT
(22) [Patent Document]
(23) (Document 001) DE Patent Publication No. 102015212869.
SUMMARY
Technical Problems
(24) The first objective of the present disclosure is to provide a dishwasher in which a means of limiting a relative displacement or a relative movement is provided integrally between an airflow guide and a connection duct part to which the airflow guide connects, such that the airflow guide is reliably supported and effectively prevented from clattering without an additional member.
(25) The second objective of the present disclosure is to provide a dishwasher in which a screw part integrally provided at a connection duct part is supported at at least three different spots at a time of a relative downward displacement or a relative downward movement of the airflow guide, such that even if quite a strong external force is applied, the airflow guide is effectively prevented from escaping from its right position.
(26) The third objective of the present disclosure is to provide a dishwasher in which a screw part integrally provided at a connection duct part is supported at at least three different spots at a time of a relative downward displacement or a relative downward movement of the airflow guide, such that the airflow guide and the connection duct part are effectively prevented from being broken since external force is divided and transferred to the connection duct part through the three different spots.
(27) The fourth objective of the present disclosure is to provide a dishwasher in which a connection duct part, extending in a way that penetrates a dry air supply hole, is fixed in the state of being spaced from the lower surface of a fastening nut fixed to a tub, such that the lower end surface of the fastening nut is exposed to a wash space, wash water is prevented from remaining between the fastening nut and the tub, and the tub and the dry air supply hole are prevented from corroding.
(28) The fifth objective of the present disclosure is to provide a dishwasher in which wash water is prevented from remaining between a fastening nut and a tub, to effectively prevent the reproduction of germs and the generation of a bad smell between the fastening nut and the tub.
(29) The sixth objective of the present disclosure is to provide a dishwasher in which a release prevention part is integrally provided to prevent an airflow guide from rotating and escaping in the state of being coupled and fixed, thereby providing dry air reliably in a predetermine direction regardless of the operation state of the airflow guide.
(30) The seventh objective of the present disclosure is to provide a dishwasher in which an airflow guide is sufficiently spaced from the edge of a tub, such that the damage and noise of the airflow guide, caused by its collision with the edge of the tub, are minimized.
(31) The eighth objective of the present disclosure is to provide a dishwasher in which an airflow guide is coupled to a dry air supply part, based on a two-stage coupling manipulation, to readily implement the setting of the right position of the airflow guide with respect to the dry air supply part and effectively prevent the misassembly of the airflow guide with respect to the dry air supply part.

(32) Aspects according to the present disclosure are not limited to the above ones, and other aspects and advantages that are not mentioned above can be clearly understood from the following description and can be more clearly understood from the embodiments set forth herein. Additionally, the aspects and advantages in the present disclosure can be realized via means and combinations thereof that are described in the appended claims.

Technical Solutions

(33) A dishwasher according to the present disclosure comprises a tub accommodating a wash target and having a wash space a front surface of which is open; a dry air supply part being disposed at a lower portion side of the tub and generating dry air for drying the wash target; an airflow guide diverting a flow direction of the dry air supplied by the dry air supply part and supplying the dry air to the wash space, wherein the dry air supply part comprises a connection duct part having an upper end that extends by penetrating a lower surface of the tub upward and supplying the dry air into the airflow guide; and a relative movement limiter limiting a rotational movement and a downward movement of the airflow guide relative to the connection duct part in a state in which the airflow guide is coupled to the connection duct part completely.

(34) The airflow guide may comprise a cylindrical duct coupling part into which the upper end of the connection duct part is inserted and coupled, and the relative downward movement of the duct coupling part may be limited by a male screw thread.

(35) The relative movement limiter may comprise at least one of protruding ribs that are formed in a way that protrudes downward toward the male screw thread at the lower end of the duct coupling part, and at a time of the relative downward movement, the lower end surface of at least one of the protruding ribs may contact the male screw thread.

(36) As the duct coupling part and the connection duct part are coupled completely, a gap having a predetermined width may be formed between the lower end surface of at least one of the protruding ribs and the male screw thread, and the relative downward movement may be limited to the predetermined width or less.

(37) At least one of the protruding ribs may comprise a first protruding rib, a second protruding rib and a third protruding rib that are arranged at regular intervals around a circular opening through which the upper end of the connection duct part passes.

(38) The heights at which the first protruding rib, the second protruding rib and the third protruding rib protrude downward from the lower end of the duct coupling part may differ respectively.

(39) Additionally, a maximum height of the first protruding rib protruding from the lower end of the duct coupling part may be a first height, a maximum height of the second protruding rib protruding from the lower end of the duct coupling part may be a second height greater than the first height, and a maximum height of the third protruding rib protruding from the lower end of the duct coupling part may be a third height greater than the second height.

(40) Further, a circumferential width of the first protruding rib, a circumferential width of the second protruding rib and a circumferential width of the third protruding rib may be the same.

(41) Further, when viewed from the upper portion side of the airflow guide, the first protruding rib, the second protruding rib and the third protruding rib may be consecutively disposed clockwise at the lower end of the connection duct part.

(42) Further, a lower end surface of the first protruding rib, a lower end surface of the second protruding rib and a lower end surface of the third protruding rib, protruding from the lower end of the duct coupling part, may be inclined surfaces the height of which respectively increases clockwise.

(43) The dry air supply part may further comprise a fastening nut that is screw-coupled to the male screw thread of the connection duct part to fix the connection duct part to the lower surface of the tub, and as the duct coupling part and the connection duct part are coupled completely, at least one of the protruding ribs may be in no contact with the fastening nut.

(44) Further, a radial maximum width of at least one of the protruding ribs may be less than a gap

between the outer circumferential surface of the connection duct part and the inner circumferential surface of the fastening nut.

(45) At least one of the protruding ribs may be inserted into the fastening nut.

(46) The duct coupling part may be coupled to the connection duct part, based on a two-stage coupling manipulation.

(47) Further, the two-stage coupling manipulation may comprise a perpendicular movement manipulation of relatively moving the airflow guide downward; and a rotational movement manipulation of relatively moving the airflow guide circumferentially after the perpendicular movement manipulation is completed.

(48) As the rotational movement manipulation starts, the gap between the lower end surface of at least one of the protruding ribs and the male screw thread may decrease gradually.

(49) Further, as the rotational movement manipulation starts, the airflow guide may rotate in a direction opposite to the direction in which the airflow guide rotates to screw-couple the fastening nut to the male screw thread of the connection duct part.

(50) Further, the relative movement limiter may further comprise a first guide groove that is formed on the inner circumferential surface of the duct coupling part and extends linearly along an up-down direction, and a second guide groove that has one end portion integrally connecting to an upper end of the first guide groove and extends along a circumferential direction in a circular arc shape, and a guide projection may be provided on the outer circumferential surface of the connection duct part, protrudes toward the inner circumferential surface of the connection duct part, and moves along the first guide groove and the second guide groove at a time of the two-stage coupling manipulation.

(51) Further, in the state in which the guide projection is inserted into the first guide groove, the perpendicular movement manipulation may be guided, and in the state in which the guide projection is inserted into the second guide groove, the rotational movement manipulation may be guided.

(52) Further, as the rotational movement manipulation is completed, a relative upward movement of the guide projection of the duct coupling part may be limited by the second guide groove.

(53) Further, in the state in which the fastening nut is screw-coupled to the connection duct part completely, the lower end surface of the fastening nut may be exposed to the wash space at least partially.

(54) The fastening nut may comprise a plurality of contact projections having an upper end that respectively connects to the lower end surface of the fastening nut integrally, and having a lower end that extends toward the lower surface of the tub.

(55) Each of the plurality of contact projections may be space from each other on the lower end surface of the fastening nut along the circumferential direction at predetermined circumferential intervals.

(56) The predetermined circumferential interval of the plurality of contact projections may be maintained regularly.

(57) The predetermined circumferential interval may be greater than a maximum circumferential thickness of the plurality of contact projections.

(58) Each of the plurality of contact projections may have the same outer shape.

(59) Each of the plurality of contact projections may have an outer shape the horizontal cross section of which decreases gradually from the upper end to the lower end.

(60) The lower end of each of the contact projections, contacting the lower surface of the tub, may be formed into a curved surface that is convex toward the lower surface of the tub.

(61) Further, a dry air supply hole may be provided on the lower surface of the tub, and the upper end of the connection duct part may pass and extend through the dry air supply hole, and a ring-type coupling surface may be formed around the dry air supply hole and pressurized by the lower end of the contact projection.

(62) The coupling surface may extend in a direction perpendicular to the direction in which the fastening nut moves while being screw-coupled to the connection duct part.
(63) The direction in which the fastening nut moves while being screw-coupled to the connection duct part may be a perpendicular direction, and the direction in which the coupling surface extends may be a horizontal direction.
(64) Further, a convergence surface having a predetermined downward inclination angle with respect to the horizontal direction may be provided around the coupling surface.
(65) Further, a cylindrical part may be provided on the lower surface of the tub, and extend circumferentially along the dry air supply hole and protrude upward toward the lower end surface of the fastening nut.
(66) The height of the cylindrical part protruding from the lower surface of the tub may remain constant along the circumferential direction.
(67) The height of the cylindrical part protruding from the lower surface of the tub may be less than the height of the contact projection protruding from the lower end surface of the fastening nut.
(68) The airflow guide may be detachably coupled to the dry air supply part without an additional coupling member.
(69) The dry air supply part may comprise a connection duct part that extends in a way that penetrates the lower surface of the tub upward, and supplies the dry air to the airflow guide, and the airflow guide may be detachably coupled to the connection duct part.
(70) The airflow guide may be coupled to the connection duct part, based on a two-stage coupling manipulation.
(71) The two-stage coupling manipulation may comprise a perpendicular movement manipulation of relatively moving the airflow guide downward.
(72) The two-stage coupling manipulation may comprise a rotational movement manipulation of relatively moving the airflow guide circumferentially after the perpendicular movement manipulation is completed.
(73) The airflow guide may comprise a duct coupling part having an upper end that is exposed to an inner flow space, and a lower end that connects to the connection duct part, and having a cylindrical shape. The upper end of the cylindrical connection duct part may be coupled to the lower end of the duct coupling part, based on the two-stage coupling manipulation.
(74) The upper end of the connection duct part may be inserted into the lower end of the duct coupling part.
(75) Further, a first guide groove and a second guide groove may be formed on the inner circumferential surface of the duct coupling part, and the first guide groove may extend linearly along the up-down direction, and the second guide groove may have one end portion integrally connecting to the upper end of the first guide groove and extending along the circumferential direction in a circular arc shape.
(76) Further, a guide projection may be provided on the outer circumferential surface of the connection duct part, and may protrude toward the inner circumferential surface of the connection duct part and join the first guide groove and the second guide groove.
(77) In the state where the guide projection joins the first guide groove, the perpendicular movement manipulation may be guided.
(78) Further, in the state where the guide projection joins the second guide groove, the rotational movement manipulation may be guided.
(79) Further, a stopper projection may be integrally provided at the second guide groove, and the stopper projection may be disposed near the other end portion of the second guide groove, and after the rotational movement manipulation is completed, block the generation of a relative rotation in a direction opposite to the direction of the rotational movement manipulation.
(80) The relative movement limiter may further comprise a release prevention part that keeps the airflow guide fixed after the rotational movement manipulation is completed, and the release

prevention part may prevent the airflow guide from rotating in a direction opposite to the direction of the rotational movement manipulation.

(81) One end portion of the release prevention part may be a fixed end portion that integrally connects to the airflow guide, and the other end portion of the release prevention part may be a free end portion that separates from the airflow guide and is elastically deformed.

(82) The dry air supply part may further comprise a fastening nut that is screw-coupled to the connection duct part and fixes the connection duct part to the tub, and at least one stopper may be integrally provided at the upper end of the fastening nut and join the free end portion of the release prevention part to prevent the airflow guide from rotating in a direction opposite to the direction of the rotational movement manipulation.

(83) The airflow guide may further comprise a release prevention part that keeps the airflow guide fixed after the rotational movement manipulation is completed.

(84) The release prevention part may prevent the airflow guide from rotating in a direction opposite to the direction of the rotational movement manipulation.

(85) The dry air supply part may comprise a connection duct part that extends in a way that penetrates the lower surface of the tub in the upward direction and supplies the dry air to the airflow guide, and the airflow guide may be detachably coupled to the connection duct part and prevented from rotating relative to the connection duct part in the opposite direction by the release prevention part.

(86) The airflow guide may comprise a lower guide which is coupled to the connection duct part and into which dry air generated in the dry air supply part is drawn; and an upper guide which is coupled to the upper side of the lower guide and has a flow space in which the dry air flows, the lower guide, comprising: a duct coupling part having an upper end that is exposed to the flow space and a lower end that is coupled to the connection duct part and having a cylindrical shape; and an edge wall that is formed in a way that surrounds the outer surface of the duct coupling part at least partially, and spaced from the duct coupling part, and the release prevention part may be integrally formed on the edge wall.

(87) The release prevention part may be formed m a way that the lower end of the edge wall is partially cut.

(88) Further, one end portion of the release prevention part may be a fixed end portion that integrally connects to the airflow guide, and the other end portion of the release prevention part may be a free end portion that separates from the airflow guide and is elastically deformed.

(89) Further, a radial thickness of the free end portion may be greater than a radial thickness of the fixed end portion.

(90) The dry air supply part may further comprise a fastening nut that is screw-coupled to the connection duct part and fixes the connection duct part to the tub, and at least one of stoppers may be integrally provided at the upper end of the fastening nut, and join the free end portion of the release prevention part and prevent the airflow guide from rotating in a direction opposite to the direction of the rotational movement manipulation.

(91) At least one of the stoppers may comprise a first stopper and a second stopper that are arranged one after another along the circumferential direction, and in the state in which the release prevention part is not elastically deformed, the free end portion may protrude at least partially further inward than the radial outer side end portion of the first stopper and the radial outer side end portion of the second stopper with respect to a radial direction.

(92) If the airflow guide relatively rotates in a direction opposite to the direction of the rotational movement manipulation, the free end portion may contact any one of one side surface of the first stopper or one side surface of the second stopper.

(93) The inner circumferential surface of the release prevention part, facing the connecting duct part, may extend toward the free end portion from the fixed end portion in the circumferential direction, and comprise a non-contact surface that is in no contact with the radial outer side end

portion of the first stopper or the radial outer side end portion of the second stopper; and a contact surface that is formed after the non-contact surface in succession, and at a time of the rotational movement manipulation, contacts the radial outer side end portion of the first stopper or the radial outer side end portion of the second stopper.
(94) The free end portion of the release prevention part may be elastically deformed based on contact with the radial outer side end portion of the first stopper or the radial outer side end portion of the second stopper.
(95) Further, a distance between the contact surface and the central axis of the duct coupling part may decrease gradually toward the free end portion.
(96) The free end portion of the release prevention part may be provided with an end portion surface that is formed in a parallel direction with the radial direction, and a tool groove may be provided on the end portion surface and concave toward the fixed end portion.
(97) Furthermore, at least one of reinforcement ribs may be integrally provided on the outer circumferential surface of the release prevention part, and protrude outward in the radial direction and extend along the circumferential direction.
Advantageous Effects
(98) A dishwasher according to the present disclosure may have the effects of evenly distributing dry air and ensuring a sufficient period for which dry air stays in a tub, thereby improving drying efficiency and reducing a drying period.
(99) The dishwasher according to the present disclosure has the effect of preventing wash water from being reversely drawn through a discharge opening that is directly exposed to a wash space.
(100) The dishwasher according to the present disclosure has the effect of preventing an airflow guide from escaping from a set position or clattering through a simple structure after the airflow guide is assembled and fixed to a dry air supply part.
(101) The dishwasher according to the present disclosure has the effect of preventing the airflow guide from escaping from its right position or being broken due to an external impact after the airflow guide is assembled and fixed to the dry air supply part.
(102) The dishwasher according to the present disclosure has the effect of preventing the corrosion of the tub and a dry air supply hole and preventing the reproduction of germs and the generation of a bad smell between the tub and a fastening nut since the lower end surface of the fastening nut is exposed to a wash space to prevent wash water from remaining between the fastening nut and the tub.
(103) The dishwasher according to the present disclosure has the effect of simplifying the processes of assembling and fixing the airflow guide to the dry air supply part since the assembly structure and fixation structure of the airflow guide spraying dry air are simplified.
(104) The dishwasher according to the present disclosure has the effect of preventing a change in a predetermined position of a discharge opening through a means of preventing the misassemble between an upper guide and a lower guide that constitute the airflow guide.
(105) The dishwasher according to the present disclosure has the effect of preventing the airflow guide from releasing or escaping from its set position through a simple structure after the airflow guide is assembled and fixed to the dry air supply part.
(106) Specific effects are described along with the above-described effects in the section of detailed description.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The accompanying drawings constitute a part of the specification, illustrate one or more embodiments in the disclosure, and together with the specification, explain the disclosure.

(2) FIG. **1** is front perspective view showing a dishwasher of one embodiment.
(3) FIG. **2** is a schematic cross-sectional view showing the dishwasher in FIG. **1**.
(4) FIG. **3** is a front perspective view showing a dry air supply part of the dishwasher of one embodiment, which is accommodated in a base.
(5) FIG. **4** is an exploded perspective view showing the dry air supply part in FIG. **3**.
(6) FIGS. **5** and **6** are front perspective views showing a bottom tub coupled to the dry air supply part in FIG. **3**.
(7) FIG. **7** is a plan view of FIG. **5**.
(8) FIG. **8** is a cross-sectional view along A-A in FIG. **7**.
(9) FIG. **9** is a plan view for describing a relative position relationship between a lower rack and a bottom tub with respect to an airflow guide of the dishwasher of one embodiment.
(10) FIG. **10** is a partial enlarged view of FIG. **9** cut in a direction parallel with the front-rear direction.
(11) FIGS. **11** and **12** are partial enlarged views of FIG. **9**.
(12) FIG. **13** is a plan view showing a relative position of a lower spray arm and an airflow guide.
(13) FIGS. **14**A to **14**C show experimental data on the distribution of measured temperatures in a drying process of the related art, and FIGS. **15**A to **15**C show experimental data on the distribution of measured temperatures in a drying process of one embodiment.
(14) FIG. **16** is an exploded perspective view showing an airflow guide and a connection duct part of the dishwasher of one embodiment.
(15) FIG. **17** is a side view showing an upper guide in FIG. **16**, and FIG. **18** is a rear perspective view showing the upper guide.
(16) FIG. **19** is a rear perspective view showing a lower guide in FIG. **16**, FIG. **20** is a front perspective view of the lower guide, and FIG. **21** is a bottom perspective view of the lower guide.
(17) FIG. **22** is a cross-sectional view showing a coupling state between the connection duct part and the lower guide.
(18) FIG. **23** is a perpendicular cross-sectional view showing a cross section in the state where the airflow guide is coupled to the connection duct part.
(19) FIG. **24** is a horizontal cross-sectional view showing a cross section in the state where the airflow guide is coupled to the connection duct part.
(20) FIGS. **25**A to **27**B are plan views and front views showing the process of assembling the airflow guide of the dishwasher of one embodiment to the connection duct part.
(21) FIGS. **28** and **29** are cross-sectional views for describing the structure and function of a release prevention part provided at the airflow guide.
(22) FIG. **30** is a partial enlarged view showing the state where the airflow guide is assembled to the connection duct part completely.
(23) FIG. **31** is a partial enlarged view showing a connection duct part to which an airflow guide is assembled.
(24) FIGS. **32** to **34** are cross-sectional views showing the cross sections of the airflow guide, the connection duct part and the bottom tub in FIG. **30**, which are cut in different positions.
(25) FIG. **35** is a cross-sectional view showing a perpendicular cross section of a fastening nut of the dishwasher of one embodiment.
(26) FIG. **36** is a perpendicular cross-sectional view showing a relationship between the fastening nut and the bottom tub in FIG. **35**.

DETAILED DESCRIPTION

(27) The above-described aspects, features and advantages are specifically described hereafter with reference to the accompanying drawings such that one having ordinary skill in the art to which the present disclosure pertains can embody the technical spirit of the disclosure easily. In the disclosure, detailed description of known technologies in relation to the disclosure is omitted if it is deemed to make the gist of the disclosure unnecessarily vague. Below, preferred embodiments

according to the disclosure are specifically described with reference to accompanying drawings. In the drawings, identical reference numerals can denote identical or similar components.
(28) The terms "first", "second" and the like are used herein only to distinguish one component from another component. Thus, the components should not be limited by the terms. Certainly, a first component can be a second component, unless stated to the contrary.
(29) Throughout the disclosure, each component can be provided as a single one or a plurality of ones, unless explicitly stated to the contrary.
(30) When one component is described as being "in the upper portion (or lower potion)" or "on (or under)" another component, one component can be directly on (or under) another component, and an additional component can be interposed between the two components.
(31) When any one component is described as being "connected". "coupled", or "connected" to another component, any one component can be directly connected or coupled to another component, but an additional component can be "interposed" between the two components or the two components can be "connected". "coupled", or "connected" by an additional component.
(32) The singular forms "a", "an" and "the" are intended to include the plural forms as well, unless explicitly indicated otherwise. It should be further understood that the terms "comprise" or "include" and the like, set forth herein, are not interpreted as necessarily including all the stated components or steps but can be interpreted as excluding some of the stated components or steps or can be interpreted as including additional components or steps.
(33) The singular forms "a", "an" and "the" are intended to include the plural forms as well, unless explicitly indicated otherwise. It should be further understood that the terms "comprise" or "include" and the like, set forth herein, are not interpreted as necessarily including all the stated components or steps but can be interpreted as excluding some of the stated components or steps or can be interpreted as including additional components or steps.
(34) Throughout the disclosure, the terms "A and/or B" as used herein can denote A, B or A and B, and the terms "C to D" can denote C or greater and D or less, unless stated to the contrary.
(35) Hereafter, the subject matter of the present disclosure is described with reference to the drawings showing the configuration of the dishwasher **1** of the embodiment.
(36) [Entire Structure of Dishwasher]
(37) Hereafter, the entire structure of the dishwasher of one embodiment is describer with reference to the accompanying drawings.
(38) FIG. **1** is a front perspective view showing a dishwasher according to the present disclosure, and FIG. **2** is a schematic cross-sectional view schematically showing the inner structure of the dishwasher according to the present disclosure.
(39) As illustrated in FIG. **1** to **2**, the dishwasher **1** according to the present disclosure comprises a case **10** forming the exterior of the dishwasher **1**, a tub **20** being installed in the case **10**, forming a wash space **21** in which a wash target is washed and having a front surface that is open, a door **30** opening and closing the open front surface of the tub **20**, a driving part **40** being disposed under the tub **20** and supplying, collecting, circulating and draining wash water for washing a wash target, a storage part **50** being provided detachably in the wash space **21** in the tub **20** and allowing a wash target to be mounted on, and a spray part **60** being installed near the storage part **50** and spraying wash water for washing a wash target.
(40) At this time, wash targets mounted in the storage part **50** may be cooking vessels such as bowls, dishes, spoons, chopsticks, and the like, and other cooking tools, for example. Hereafter, the wash targets are referred to as cooking vessels, unless mentioned otherwise.
(41) The tub **20** may be formed into a box the front surface of which is open entirely, and may be a so-called tub.
(42) The tub **20** may have a wash space **21** therein, and its open front surface may be opened and closed by the door **30**.
(43) The tub **20** may be formed in a way that a metallic sheet having strong resistance against high-

temperature and moisture, e.g., a stainless steel-based sheet, is pressed.
(44) Additionally, a plurality of brackets may be disposed on the inner surfaces of the tub **20** and allow functional components such as a storage part **50**, a spray part **60** and the like, which are described below, to be supported and installed in the tub **20**.
(45) The driving part **40** may comprise a sump **41** storing wash water, a sump cover **42** distinguishing the sump **41** from the tub **20**, a water supply part **43** supplying wash water to the sump **41** from the outside, a drain part **44** discharging wash water of the sump **41** to the outside, and a water supply pump **45** and a supply channel **46** for supplying wash water of the sump **41** to the spray part **60**.
(46) The sump cover **42** may be disposed at the upper side of the sump **41**, and distinguish the sump **41** from the tub **20**. Additionally, the sump cover **42** may be provided with a plurality of return holes for returning wash water, having sprayed to the wash space **21** through the spray part **60**, to the sump **41**.
(47) That is, wash water having sprayed toward cooking vessels from the spray part **60** may fall to the lower portion of the wash space **21** and return to the sump **41** through the sump cover **42**.
(48) The water supply pump **45** is provided in a lateral portion or the lower portion of the sump **41**, and pressurizes wash water and supplies the same to the spray part **60**.
(49) One end of the water supply pump **45** may connect to the sump **41**, and the other end may connect to the supply channel **46**. The water supply pump **45** may have an impeller **451**, a motor **453** and the like, therein. As power is supplied to the motor **453**, the impeller **451** may rotate, and wash water of the sump **41** may be pressurized and then supplied to the spray part **60** through the supply channel **46**.
(50) The supply channel **46** may selectively supply the wash water supplied by the water supply pump **45** to the spray part **60**.
(51) For example, the supply channel **46** may comprise a first supply channel **461** connecting to a lower spray arm **61**, and a second supply channel **463** connecting to an upper spray arm **62** and a top nozzle **63**. The supply channel **46** may be provided with a supply channel diverting valve **465** selectively opening and closing the supply channels **461**, **463**.
(52) At this time, the supply channel diverting valve **465** may be controlled to allow each of the supply channels **461**, **463** to be opened consecutively or opened simultaneously.
(53) The spray part **60** is provided to spray wash water to cooking vessels and the like stored in the storage part **50**.
(54) Specifically, the spray part **60** may comprise a lower spray arm **61** being disposed under the tub **20** and spraying wash water to a lower rack **51**, an upper spray arm **62** being disposed between the lower rack **51** and an upper rack **52** and spraying wash water to the lower rack **51** and the upper rack **52**, and a top nozzle **63** being disposed in the upper portion of the tub **20** and spraying wash water to a top rack **53** or the upper rack **52**.
(55) In particular, the lower spray arm **61** and the upper spray arm **62** may be provided in the wash space **21** of the tub **20**, and spray wash water toward cooking vessels in the storage part **50** while rotating.
(56) The lower spray arm **61** may be rotatably supported at the upper side of the sump cover **42** such that the lower spray arm **61** may spray wash water to the lower rack **51** while rotating under the lower rack **51**.
(57) Additionally, the upper spray arm **62** may be rotatably supported by a spray arm holder **467** such that the upper spray arm **62** may spray wash water while rotating between the lower rack **51** and the upper rack **52**.
(58) The tub **20** may be further provided with a reflection plate on a lower surface **25** thereof, to enhance washing efficiency, and the reflection plate diverts the direction of wash water having sprayed from the lower spray arm **61** to an upward direction (U-direction).
(59) Since a well-known configuration can be applied to the configuration of the spray part **60**,

detailed description of the configuration of the spray part **60** is omitted hereafter.
(60) The storage part **50** for storing cooking vessels may be provided in the wash space **21**.
(61) The storage part **50** may be withdrawn through the open front surface of the tub **20** from the inside of the tub **20**.
(62) For example. FIG. **2** shows an embodiment provided with a storage part comprising a lower rack **51** that is disposed in the lower portion of the tub **20** and stores relatively large-sized cooking vessels, an upper rack **52** that is disposed at the upper side of the lower rack **51** and stores medium-sized cooking vessels, and a top rack **53** that is disposed in the upper portion of the tub **20** and stores small-sized cooking vessels and the like. However, the subject matter of the present disclosure is not limited to the embodiment. Hereafter, a dishwasher that is provided with three storage parts **50**, as illustrated, is described.
(63) Each of the lower rack **51**, the upper rack **52** and the top rack **53** may be withdrawn outward through the open front surface of the tub **20**.
(64) To this end, the tub **20** may have a guide rail **54**, on both lateral walls thereof that form the inner circumferential surface of the tub **20**, and for example, the guide rail **54** may comprise an upper rail **541**, a lower rail **542**, a top rail **543** and the like.
(65) Each of the lower rack **51**, the upper rack **52** and the top rack **53** may be provided thereunder with wheels. A user may withdraw the lower rack **51**, the upper rack **52** and the top rack **53** outward through the front surface of the tub **20** to easily store cooking vessels on the racks or take out cooking vessels from the racks after a washing process.
(66) The guide rail **54** may be provided as a fixed guide rail that guides the withdrawal and insertion of the spray part **60** in the form of a simple rail or as a stretchable guide rail which guides the withdrawal and storage of the spray part **60** and the withdrawal distance of which increases as the spray part **60** is withdrawn.
(67) The door **30** is used for opening and closing the open front surface of the tub **20** that is described above.
(68) In some examples, a hinge part for opening and closing the door **30** is provided in the lower portion of the open front surface, and the door **30** is open with respect to the hinge part as a rotation axis.
(69) The door **30** may be provided with a handle **31** and a control panel **32** on the outer surface thereof. The handle **31** is used for opening the door **30**, and the control panel **32** is used for controlling the dishwasher **1**.
(70) As illustrated, the control panel **32** may be provided with a display **33** that visually displays information on a current operation state and the like of the dishwasher, and a button part **34** comprising a selection button to which the user's selection manipulation is input, a power button to which the user's manipulation for turning on-off the power source of the dishwasher is input, and the like.
(71) The inner surface of the door **30** may form a mounting surface that supports the lower rack **51** of the storage part **50** as the door **30** is opened as well as forming one surface of the tub **20** as the door **30** is closed.
(72) To this end, as the door **30** is fully opened, the inner surface of the door **30** forms a horizontal surface in the same direction where the guide rail **54**, by which the lower rack **51** is guided, extends, for example.
(73) The door **30** may be provided rotatably between a closed position and a fully-open position, and an intermediate still position may be formed between the closed position and the fully open position.
(74) The door **30** may stand still in the intermediate still position, and at this time, the wash space **21** of the tub **20** may be open outward partially. When the door **30** is disposed at the intermediate still position, a dry air supply part **80** described hereafter may operate to supply high-temperature dry air or low-temperature dry air to the wash space **21**.

(75) Though not illustrated, the dry air supply part **80** may be provided in the lower portion of the tub **20** and generate high-temperature dry air and supply the high-temperature dry air into the tub **20**.
(76) Hereafter, a detailed configuration of the dry air supply part **80** is described with reference to FIG. **3**
(77) [Detailed Configuration of Dry Air Supply Part]
(78) Hereafter, the detained configuration of the above-described dry air supply part **80** is described with reference to FIGS. **3** to **7**.
(79) As illustrated in FIG. **3**, the dry air supply part **80** may be accommodated in a base **90** and may be disposed to be supported by a lower surface **91** of the base **90**.
(80) For example, the dry air supply part **80** may be disposed in a position adjacent to a rear surface **93** of the base **90**, and disposed in a position between a leakage detecting part and the rear surface **93** of the base **90**, approximately in parallel with the rear surface **93** of the base **90**.
(81) The position in which the dry air supply part is disposed may be selected considering the characteristics of the dry air supply part **80** that generates heat of about 100° C. or greater in a high-temperature dry air supply mode. That is, the dry air supply part may be disposed to avoid electronic components that are greatly affected by high-temperature heat.
(82) Additionally, the arrangement position of the dry air supply part may be selected based on the position of a dry air supply hole **254** formed on the lower surface **25** of the tub **20**. That is, considering the user's safety, the dry air supply hole **254** into which dry air flows may be formed at the corner of the lower surface **25** of the tub **20**, which is adjacent to the rear surface and the left side surface of the tub **20**.
(83) For the dry air supply part **80** to effectively generate dry air and supply the same to the dry air supply hole **254** formed in the above-described position, the dry air supply part **80** may be disposed at the lower side of the dry air supply hole **254**.
(84) The arrangement position of the dry air supply part **80** is described exemplarily. The dry air supply part **80** may be disposed near a left side surface **94**, a right side surface **95** or a front surface **92** of the base **90** rather than the rear surface **93** of the base **90**. Hereafter, the dry air supply part **80** disposed near the rear surface **93** of the base **90** approximately in parallel with the rear surface **93** is described, but the position of the dry air supply part **80** is not limited.
(85) Additionally, a support rib for supporting the dry air supply part **80** and preventing the escape of the dry air supply part **80**, a plurality of guide ribs setting the position of a leakage detecting part that detects whether wash water leaks from the tub **20** and preventing the escape of the leakage detecting part, and a wash water rib for guiding wash water being discharged from the dry air supply part **80** to the leakage detecting part may be provided on the lower surface **91** of the base **90**.
(86) The support rib, the guide ribs and the wash water rib may be formed integrally on the lower surface **91** of the base **90**, for example.
(87) A first leg, a second leg **892** and a third leg **893** of the dry air supply part **80** described hereafter may be coupled to the support rib, based on a non-fastening method. That is, the first leg, the second leg **892** and the third leg **893** may be simply held at the support rib without an additional fastening means such that the dry air supply part **80** may be supported in up-down, front-rear and left-right directions.
(88) FIG. **4** shows a detailed configuration of the dry air supply part **80**.
(89) As illustrated, the dry air supply part **80** generating dry air and supplying the same into the tub **20** may comprise an air blowing fan that generates dry airflow F to be supplied into the tub **20**, a heater that heats dry air, a heater housing **81** that has an air passage in which the heater is accommodated, and a filtering part **88** that filters air to be suctioned into the air blowing fan.
(90) The air blowing fan is disposed at the upstream side in the direction of dry airflow F with respect to the heater and the heater housing **81**, and accelerates air to the air passage formed in the heater housing **81** to generate dry airflow F.

(91) The air blowing fan, and an air blowing motor generating rotational driving force of the air blowing fan may be mutually modularized, and form an assembly in a way that the air blowing fan and the air blowing motor are accommodated in a fan housing **82**.
(92) The air blowing fan and the fan housing **82** may be fixed to a housing connector **87** that connects a filer housing **881** of the below-described filtering part **88** and the heater housing **81**.
(93) Specifically, the air blowing fan and the fan housing **82** may be accommodated entirely in the filter housing **881** in the state of being fixed to a connection tab **872** the housing connector **87**.
(94) The type of the air blowing fan to be applied to the dry air supply part **80** is not limited, but a sirocco fan, for example, is preferred considering the position and space limitations in the installation of the air blowing fan.
(95) When a sirocco fan is applied as illustrated, filtered air may be suctioned from a lower surface of the fan housing **82**, in a direction parallel a direction from the center of the sirocco fan to the rotational axis of the same, and be accelerated and discharged outward in the radial direction.
(96) The accelerated and discharged air may form dry airflow F and be drawn into the air passage in the heater housing **81** through the fan housing **82** and an inlet **8712** of the housing connector **97**.
(97) At this time, the air blowing fan, e.g., a sirocco fan, and a rotation shaft **8251** of the motor may be disposed to have directionality approximately parallel with the up-down direction (U-D direction), and filtered air may be suctioned through the lower surface of the fan housing **82**, for example.
(98) Further, a PCB substrate for controlling the moor may be built into an upper surface **821** of the fan housing **82**, which corresponds to the opposite side of the lower surface into which filtered air is suctioned.
(99) The fan housing **82**, as illustrated, may be fixed to a ring-type connection tab **872** provided at the housing connector **87** through a fastening means such as a screw bolt that is not illustrated, and the like, for example.
(100) The connection tab **872** may be provided with a pair of fastening bosses that extend from the upper surface of the connection tab **872** in the upward direction (U-direction).
(101) To support the fan housing **82** and the heater housing **81**, the first leg protruding toward the base may be integrally formed under the housing connector **87**.
(102) The heater may be indirectly supported in the state of separating from the heater housing **81** and a connector main body **871**.
(103) The front end side of the heater may be supported by a terminal fixation part, in the state of separating from the housing connector **87**. A c may be fixed to the front surface of the terminal fixation part, in the state of protruding outward.
(104) An entirely-open rear end portion **871***b* of the housing connector **87** may be fixed while being fitted and coupled to the heater housing **81**.
(105) The heater is disposed in the air passage formed in the heater housing **81**, and preferably, is directly exposed to dry airflow F in the air passage and heats the dry airflow F.
(106) When the dry air supply part **80** supplies high-temperature dry air, power may be supplied to the heater, and the heater may heat dry air, and when the dry air supply part **80** supplies low-temperature dry air, the supply of power to the heater may be cut off, and the heater may stop operating.
(107) At this time, when low-temperature dry air is supplied, the air blowing motor may keep operating to generate dry airflow F.
(108) The type of the heater provided in the dry air supply part **80** of one embodiment is not limited, but a tube-type sheath heater may be selected since the sheath heater has a relatively simple structure, ensures excellent heat generation efficiency and helps to prevent electric leakage caused by the reverse inflow of wash water that comes in from the tub **20** reversely, for example.
(109) To enhance heat exchange efficiency, the heater that is a sheath heater may have a stereoscopic shape with a plurality of bends, to be directly exposed to dry airflow F at the air

passage in the heater housing **81** and ensure a maximum heat transfer surface.
(110) Additionally, a pair of terminals for receiving power may be formed in one end portion and the other end portion of the heater.
(111) The rear end side of the heater may be fixed and supported by a single heater bracket **845** disposed in the heater housing **81**. That is, the rear end side of the heater may be supported on the air passage through the heater bracket **845** in the state of being separated from the heater housing **81**.
(112) Further, a temperature sensor as a temperature sensing part **86** sensing the temperature of high-temperature dry air generated through the heater or detecting the overheating of the heater may be provided on the upper side surface of the heater housing **81**.
(113) For example, the temperature sensor may comprise a thermistor that senses the temperature of dry air, and a thermostat that detects the overheating of the heater.
(114) An output signal of the temperature sensor may be delivered to a non-illustrated controller, and the controller may receive the output signal of the temperature sensor to determine the temperature of high-temperature dry air and the overheating of the heater. As the heater overheats, the controller may cut off the supply of power to the heater and change the operation mode of the dry air supply part **80** from the high-temperature dry air supply mode to the low-temperature dry air supply mode.
(115) The heater housing **81** may be formed into a hollow hole that has a vacant inner space such that the air passage, in which the above-described heater and heater bracket **845** are disposed, is formed.
(116) At this time, for dry airflow F to move, the front end portion of the heater housing **81**, corresponding to the upper stream side with respect to the direction of the movement of the dry airflow F. and the rear end portion of the heater housing **81**, corresponding to the lower stream side with respect to the direction of the movement of the dry airflow F, may be open at least partially.
(117) The dry air supply part **80** may further comprise a connection duct part **85** that is coupled to an outlet, formed at the left end side of the heater housing **81** and being open in the upward direction (U-direction), and has an air passage therein.
(118) As described above, the heater housing **81** and the air blowing fan are disposed under the lower surface **25** of the tub **20**, e.g., a bottom tub **20***c*. The connection duct part **85** guides dry air being discharged from the heater housing **81** to a predetermined position, i.e., the dry air supply hole **254** formed at the tub **20**.
(119) For example, the predetermined position may be the lower surface **25** of the tub **20**, and the dry air supply hole **254** into which dry airflow F guided to the connection duct part **85** is drawn may be formed at a corner of the lower surface **25** of the tub **20**, which is adjacent to a rear surface **23** and a left side surface **26**.
(120) As shown in the illustrative embodiment, a duct main body **851** of the connection duct part **85** may have a shape that is capable of changing the direction of dry airflow and connecting the dry air supply hole **254** of the tub **20** and the outlet of the heater housing **81**.
(121) For example, the duct main body **851** of the connection duct part **85** may have a cylinder shape that allows of the fluid communication of a lower end portion **8512** with the outlet of the heater housing **81** and allows an upper end **8511** to extend in the upward direction (U-direction) and connect to the dry air supply hole **254**.
(122) The lower end portion **8512** of the duct main body **851** may be coupled the heater housing **81** in a sliding manner.
(123) Further, considering the cross section of the rectangle-shaped outlet of the heater housing **81**, the lower end portion of the duct main body **851** may have a rectangle pillar shape, and for the prevention of leakage, the upper end **8511** of the duct main body **851** may have a cylinder shape.
(124) That is, the duct main body **851** may have a cylinder shape to improve the efficiency of a coupling between the upper end **8511** of the duct main body **851** and the dry air supply hole **254** of

the tub **20** and to prevent leakage.
(125) An airflow guide **83** may be coupled to the upper end **8511** of the duct main body **851** and divert the direction of dry airflow being supplied through the duct main body **851** to supply the dry airflow to the wash space.
(126) The filtering part **88** may be disposed in the upper stream of the heater with respect to the direction of the flow of dry airflow, to filter air to be suctioned into the air blowing fan and supply the filtered air to the heater.
(127) Specifically, the filtering part **88** may comprise a filter member **883** that filters air to be suctioned into the air blowing fan, and a hollow hole-type filter housing **881** that has a filter accommodation space S**1** in which the filter member **883** is disposed in a replaceable manner and a fan housing accommodation space S**2** in which the fan housing **82** is disposed.
(128) As illustrated in FIG. **4**, the filter housing **881** may comprise a first housing **8811** and a second housing **8812** that are disposed in the form of a segment body that is segmented with respect to the up-down direction (U-D direction), for example. At this time, the first housing **8811** may be the upper housing, and the second housing **8812** may be the lower housing.
(129) The filter housing **881** accommodates and supports the filter member **883** and the fan housing **82** of the air blowing fan.
(130) Accordingly, the first housing **8811** may be divided into a filter accommodation part **8811***a* and a fan housing accommodation part **8811***b* such that the first housing **8811** accommodates and supports the filter member **883** and the fan housing **82** at least partially, preferably, accommodates and supports the upper portion of the filter member **883** and the upper portion of the fan housing **82**.
(131) As illustrated, the lower surfaces of the filter accommodation part **8811***a* and the fan housing accommodation part **8811***b* of the first housing **8811** are open entirely to allow the second housing **8812** to be coupled to the lower sides of the filter accommodation part **8811***a* and the fan housing accommodation part **8811***b* of the first housing **8811**.
(132) The filter accommodation part **8811***a* may be formed further upstream than the fan housing accommodation part **8811***b* with respect to the direction of the flow of dry airflow, and in the illustrative embodiment, formed on the right of the fan housing accommodation part **8811***b*.
(133) The filter accommodation part **8811***a*, for example, may have an outer shape of a partial cylinder to accommodate the filter member **883** having a cylinder shape in a way that the filter member **883** may be inserted and withdrawn when the filter member **883** is replaced.
(134) Additionally, a filter guide rib may be integrally provided in the filter accommodation part **8811***a* and have a shape similar to that of a filter guide rib **8812***f* of the second housing **8812** described hereafter.
(135) The filter accommodation part **8811***a* may have a coupling opening **8811***c* at the upper end thereof, and the coupling opening **8811***c* is open in the form of a circle in response to the outer shape of the filter member **883**. The filter member **883** may move downward through the coupling opening **8811***c*, and move to a filter accommodation part **8812***a* of the second housing **8812**.
(136) The fan housing accommodation part **8811***b* may be formed further downstream than the filter accommodation part **8811***a* with respect to the direction of the flow of dry airflow, and in the illustrative embodiment, formed integrally at the filter accommodation part **8811***a*, on the right of the filter accommodation part **8811***a*, near the heater housing **81**.
(137) The fan housing accommodation part **8811***b* may have an inner shape corresponding to the outer shape of the upper portion of the fan housing **82**, to cover the upper portion of the air blowing fan entirely. For example, the fan housing accommodation part **8811***b* may have an upper surface formed into a flat plate.
(138) As illustrated, the upper surface of the first housing **8811** may have an inclined surface **8811***b***1** that connects the upper end of the filter accommodation part **8811***a* and the fan housing accommodation part **8811***b*.

(139) The second housing **8812** of the filter housing **881** is coupled to the lower portion of the first housing **8811** and forms a sealed accommodation space, and accommodates and supports the lower portions of the filter member **883** and the fan housing **82**.
(140) Like the first housing **8811**, the second housing **8812** may be divided into a filter accommodation part **8812***a* and a fan housing accommodation part **8812***b*, to accommodate and support the lower portion of the filter member **883** and the lower portion of the fan housing **82**.
(141) As illustrated, the upper end of the second housing **8812** may be open entirely to be coupled to the lower end of the first housing **8811**.
(142) In response to the filter accommodation part **8811***a* of the first housing **8811**, the filter accommodation part **8812***a* of the second housing **8812**, provided under the filter accommodation part **8811***a* of the first housing **8811**, may be provided with a plurality of filter guide ribs that guides the filter member **883**'s movement and prevents the filter member **883**'s escape from the right position at a time of inserting the filter member **883**.
(143) Additionally, in response to the filter member **883**'s outer shape formed into a cylinder, the plurality of filter guide ribs may be arranged and disposed radially around the filter member **883**.
(144) As the center of the plurality of filter guide ribs **8812***f*, a lower suction opening **8812***c* may be formed on the bottom surface of the filter accommodation part **8812***a* in a penetrating manner, and is open toward the lower surface of the base **90** and allows external air to be suctioned.
(145) The lower suction opening **8812***c* may have a circle shape to correspond to the shape of a lower opening of the filter member **883** having a cylinder shape, and a relative position and size of the lower suction opening **8812***c* may be determined to allow external air to pass through the lower opening and to be smoothly suctioned into the filter member **883**.
(146) Additionally, as one airtight means, a pair of ring-type ribs may be formed around the lower suction opening of the bottom surface of the second housing **8812**, and prevent non-filtered external air to be leaked and suctioned into the inner space of the filter housing **881** directly.
(147) The fan housing accommodation part **8812***b* may be formed further downstream than the filter accommodation part **8812***a* with respect to the direction of the flow of dry airflow, and in the illustrative embodiment, formed integrally at the filter accommodation part **8812***a*, on the right of the filter accommodation part **8812***a*, near the heater housing **81**.
(148) The fan housing accommodation part **8811***b* may have an inner shape corresponding to the outer shape of the lower portion of the fan housing **82**, to cover the lower portion of the air blowing fan entirely.
(149) The bottom surface of the fan housing accommodation part **8811***b* may be spaced a predetermined distance apart from the lower surface of the fan housing **82**, to allow filtered air to be suctioned effectively, and for example, be formed into a flat surface in a direction parallel with the horizontal direction.
(150) As a means of spacing the fan housing **82** apart from the bottom surface of the fan housing accommodation part **8811***b* and supporting the fan housing **82**, a plurality of uplifted surface parts and a screw boss that protrude from the bottom surface may be provided in the fan housing accommodation part **8812***b*.
(151) In the first housing **8811** and the second housing **8812** that are disposed in the form of a segment body as described above, the lower end of the first housing **8811** and the upper end of the second housing **8812** may be detachably coupled to each other.
(152) To achieve the above-described detachable coupling relationship, a fastening tab **8811***d* extending toward the second housing **8812** is provided at the lower end of the first housing **8811**, and a hook projection **8812***d* may be provided at the upper end of the second housing **8812** and fastened to the fastening tab **8811***d* based on a hook coupling.
(153) A tub connection duct **882** may be detachably coupled and fastened to the coupling opening **8811***c* of the filter accommodation part **8811***a* of the first housing **8811**.
(154) The filter member **883** may be replaced through a lower surface **25** of the tub **20**.

(155) To this end, the filter accommodation part **8811***a* of the first housing **8811** needs to connect to the lower surface **25** of the tub **20**, and the tub connection duct **882** connects the lower surface **25** of the tub **20** and the filter accommodation part **8811***a* of the first housing **8811**.
(156) The tub connection duct **882** may be integrally provided at the filter accommodation part **8811***a* of the first housing **8811**. The tub connection duct **882** may be provided additionally in the first housing **8811**, as illustrated.
(157) Like the duct main body **851** of the above-described connection duct part **85**, an upper end portion **8821** of the tub connection duct **882** may pass through the lower surface **25** of the tub **20** and extend upward.
(158) A filter replacement hole **253** (FIG. **11**) may be provided on the lower surface **25** of the tub **20** to allow the upper end portion **8821** of the tub connection duct **882** to be inserted.
(159) A sump hole **252** on which a sump **41** is mounted may be provided in the central portion of the lower surface **25** of the tub **20**. The lower surface **25** of the tub **20** may have a convergence surface having an inclination angle at which the convergence surface gradually inclines downward toward the sump hole **252**, to allow wash water to be effectively converged on the sump hole **252**.
(160) As illustrated, the filter replacement hole **253** may be formed on the convergence surface, at the rear of the sump hole **252**.
(161) To distinguish the filter replacement hole **253** from the dry air supply hole **254**, the filter replacement hole **253** may be formed at the corner adjacent to the rear surface and the right side surface, on the lower surface **25** of the tub **20**. Additionally, to easily insert and withdraw the filter member **883** for replacement, the filter replacement hole **253** may be disposed closer to the front surface of the tub **20** than the dry air supply hole **254** and disposed further rearward than a water softener communication hole **255**.
(162) The water softener communication hole **255** formed in front of the filter replacement hole **253**, for example, may be used to insert a water softening agent into a water softener provided under the water softener communication hole **255**, and the like, or used for the replacement and maintenance and repairs of another component such as a purification filter of a water supply part, and the like.
(163) The filter replacement hole **253** may be disposed between the water softener communication hole **255** and the dry air supply hole **254** with respect to the front-rear direction or the left-right direction.
(164) That is, the filter replacement hole **253** may be disposed outside a virtual extension line that connects the water softener communication hole **255** and the dry air supply hole **254**.
(165) By doing so, even if the lower surface **25** of the tub **20** has a plurality of openings, the strength, torsional rigidity and flexural rigidity of the tub **20** may not decrease.
(166) Additionally, to distinguish the filter replacement hole **253** from the water softener communication hole **255** formed in front of the filter replacement hole **253**, a sealing cap **884** having a different shape or color from the water softener communication hole **255** may be applied at the upper end of the tub connection duct **882** that passes through the filter replacement hole **253** and is exposed to the wash space.
(167) As described above, the filter replacement hole **253** is provided on the convergence surface provided on the lower surface **25** of the tub **20**. Thus, the tub connection duct **882**'s upper end portion and flange coupled to the filter replacement hole **253** may have a predetermined inclination angle with respect to the perpendicular direction in response to the inclination angle of the convergence surface of the tub **20**, i.e., may be formed to incline with respect to the perpendicular direction.
(168) A first gasket **885** may be further provided between the flange **8823** of the tub connection duct **882** and the lower surface **25** of the tub **20**, to prevent a fastening nut **886** from loosening and prevent leakage.
(169) As the tub connection duct **882** is fixed to the lower surface **25** of the tub **20** through the

fastening nut **886**, the sealing cap **884** may be coupled to the upper end portion **8821** of the tub connection duct **882** exposed to the inside of the tub **20**. At this time, an airtight ring **887** for preventing leakage may be disposed between the sealing cap **884** and the upper end portion **8821** of the tub connection duct **882**.
(170) Additionally, an upper suction opening **8826** into which external air is suctioned may be formed under the flange corresponding to the upper side of the filter accommodation space S**1**, between the upper end portion and a lower end portion of the tub connection duct **882**, in a penetrating manner.
(171) The upper suction opening **8826** may be formed in a way that penetrates the cylinder-type tub connection duct **882** from the inner circumferential surface thereof to the outer circumferential surface thereof. For example, the upper suction opening **8826** may be provided as a plurality of penetration openings arranged and formed along the circumferential direction of the tub connection duct **882**.
(172) The upper suction opening **8826** may be formed higher than an upper opening of the filter member **883**, in the state where the filter member **883** is disposed in the filter accommodation space S**1**. Accordingly, the upper suction opening **8826** may be formed between the tub **20** and the upper surface of the filter member **883** with respect to the up-down direction.
(173) After external air having passed through the upper suction opening **8826** in a direction parallel with the horizontal direction enters into the filter member **883**, the direction of the airflow changes, and the external air may be filtered while passing through the outer circumferential surface of the filter member **883**.
(174) The suction path of external air and the flow path of dry airflow having passed through the filter member **883** are described hereafter with reference to FIG. **8**.
(175) [Air Flow Path Before and After Filtering]
(176) Hereafter, the flow path of external air before the external air passes through the filter member **883** of the dishwasher **1** of one embodiment, and the flow path of dry airflow F after external air passes through the filter member **883** and is filtered are described with reference to FIG. **8**.
(177) The first housing **8811** and the second housing **8812** of the filter housing **881** of the dishwasher **1** of one embodiment are spaced from each other in the up-down direction, and in the filter housing **881**, external air is suctioned through a plurality of suction openings that are open toward a space between the base **90** and the tub **20**.
(178) As described above, the plurality of suction openings may comprise the upper suction opening **8826** provided at the tub connection duct **882**, at the upper side of the filter accommodation space S**1**, and the lower suction opening **8812***c* provided on the bottom surface of the second housing **8812**, at the lower side of the filter accommodation space S**1**.
(179) As described above, the upper suction opening **8826** and the lower suction opening **8812***c* are spaced from each other and disposed respectively in the uppermost position and the lowermost position of the filter housing **881** with respect to the space between the tub **20** and the base **90**. Accordingly, in the state where the effect of the flow rate of air suctioned respectively into the upper suction opening and the lower suction opening is minimized, external air may flow into the filter housing **881** through the two suction openings, thereby ensuring more flow rate of air required to dry a wash target and spending less time drying a wash target than usual.
(180) As illustrated, the upper suction opening **8826** is open in a direction approximately parallel with the horizontal direction. Accordingly, external air suctioned into the upper suction opening **8826** forms airflow of a direction parallel with the horizontal direction.
(181) The lower suction opening **8812***c* is formed on the bottom surface that extends horizontally. Accordingly, the lower suction opening **8812***c* is open toward the base **90** in a direction parallel with the perpendicular direction, and external air suctioned into the lower suction opening **8812***c* forms airflow of a direction parallel with the perpendicular direction.

(182) External air suctioned through the upper suction opening **8826** may enter into the upper opening of the filter member **883** disposed right under the upper suction opening **8826** in the state where the filter member **883** is disposed in the filter accommodation space S**1**.
(183) Additionally, external air suctioned through the lower suction opening **8812***c* may enter into the lower opening of the filter member **883** disposed right on the lower suction opening **8812***c* in the state where the filter member **883** is disposed in the filter accommodation space S**1**.
(184) In the state where the filter member **883** is disposed, an airtight means of preventing non-filtered air from being suctioned into the filter housing **881** may be provided at the upper end side and the lower end side of the filter member **883**.
(185) Thus, external air suctioned into the upper suction opening **8826** and the lower suction opening **8812***c* may enter respectively into the upper opening and the lower opening of the filter member **883**, without leaking.
(186) Further, in the state where the filter member **883** is disposed in the filter accommodation space S**1**, the upper opening of the filter member **883** is open toward the lower surface **25** of the tub **20**, and the lower opening of the filter member **883** is open toward the lower surface of the base **90**. Accordingly, the direction of airflow of external air changes downward while passing through the upper opening, and external air having passed through the lower opening flows upward.
(187) As described above, external air suctioned into the filter member **883** may pass through the filtering material of the filter member **883** and be evenly suctioned entirely in the up-down direction and circumferential direction.
(188) Further, external air suctioned into the inner circumferential surface of the filter member **883** is filtered, and while passing through the outer circumferential surface of the filter member **883**, is discharged, and immediately after the discharge, the direction of the flow of the external air changes.
(189) As illustrated in FIG. **8**, the direction of the flow of the filtered air having passed through the outer circumferential surface of the filter member **883** may change toward the lower surface of the fan housing **82** that is open toward the bottom surface of the filter housing **881**.
(190) The lower surface of the fan housing **82** is disposed in a position spaced upward from the bottom surface, between the lower end and the upper end of the filter member **883**. Accordingly, air having passed through the filter member **883** in a position higher than the lower surface of the fan housing **82** flows downward to the lower surface of the fan housing **82**, and air having passed through the filter member **883** in a position lower than the lower surface of the fan housing **82** flows upward to the lower surface of the fan housing **82**.
(191) Filtered air drawn into the fan housing **82** through the above-described flow path is accelerated by the air blowing fan and then drawn into the housing connector **87** and the inner space of the heater housing **81**, such that dry airflow F is formed.
(192) [Detailed Configuration of Airflow Guide]
(193) Hereafter, a detailed configuration of the airflow guide **83** of the dishwasher **1** of one embodiment is described with reference to FIGS. **1** to **9**.
(194) As illustrated in FIG. **9**, the airflow guide **83** may be disposed between the lower surface **25** of the bottom tub **20***c* and the lower rack **51**, near the lower surface **25** of the bottom tub **20***c*, and divert the direction of the flow of dry airflow F supplied through the duct main body **851**.
(195) Specifically, the airflow guide **83** may be disposed near the corner formed between the left side surface **26** and the rear surface **23** of the bottom tub **20***c* or near the corner formed between the right side surface **27** and the rear surface **23** of the bottom tub **20***c*. In response to the position of the airflow guide **83**, the above-described dry air supply hole **254** for transferring dry air may be formed on the lower surface **25** of the bottom tub **20***c*.
(196) FIGS. **1** to **9** show that the airflow guide **83** and the dry air supply hole **254** are adjacent to the lower surface **25** of the bottom tub **20***c*, near the corner formed between the left side surface **26** and the rear surface **23** of the bottom tub **20***c*, for example. Hereafter, for convenience, the airflow

guide **83** and the dry air supply hole **254**, which are disposed near the corner formed between the left side surface **26** and the rear surface **23** of the bottom tub **20***c* as illustrated, are described as an example, but not limited.
(197) The corner formed between the left side surface **26** and the rear surface **23** of the bottom tub **20***c* corresponds to a position farthest from the upper end of the front surface **22** of the tub **20** that is partially open in the drying stage.
(198) Thus, the period for which dry air sprayed from the airflow guide **83** remains in the tub **20** may extend effectively. By doing so, dry air may be supplied to the lower rack **51**, the upper rack **52** and the top rack **53** evenly and then discharged through the upper end of the front surface of the tub, enabling thermal energy of the dry air to be transferred to wash targets effectively and significantly promoting the effect of drying the wash targets.
(199) Additionally, since the airflow guide **83** is spaced a predetermined distance apart from the rear surface **23**, the lower surface **25** and the left side surface **26** of the bottom tub **20***c*, food and the like may be effectively prevented from being fitted and fixed between the airflow guide **83** and the bottom tub **20***c*.
(200) The period for which dry air remains in the tub **20** may further increase, based on the adjustment of the direction of the spray of dry air from the airflow guide **83**.
(201) That is, a discharge opening **833** of the airflow guide **83**, from which dry air sprays, may be formed in a position where dry air does not directly spray toward the lower rack **51** and wash targets stored on the lower rack **51**.
(202) Specifically, the airflow guide **83** of the dishwasher **1** of one embodiment may discharge dry air in a direction that is not the upward direction (U-direction) perpendicular to the lower surface **25** of the bottom tub **20***c* or the direction in which dry air does not spray directly toward the lower rack **51**.
(203) To this end, the discharge opening **833** from which dry air sprays may be formed on the right side surface of the airflow guide **83** to discharge dry air in a direction approximately parallel with the rear surface **23** of the bottom tub **20***c*.
(204) As illustrated in FIGS. **9** and **10**, the front-rear (F-R direction) width of the airflow guide **83** may be greater than the left-right (Le-Ri direction) width in the state where the airflow guide **83** is coupled to the duct main body **851** of the connection duct part **85**, and the discharge opening **833** may be formed continuously throughout the right side surface of the airflow guide **83**, facing the right side surface **27** of the bottom tub **20***c*, and the rear surface of the airflow guide **83**. That is, the discharge opening **833** of the airflow guide **83** may have the directionality that does not face the door directly and face the front surface **22** of the tub **22** or the door **30** linearly.
(205) At this time, the discharge opening **833** of the airflow guide **83** may be formed into a slit or have an oblong shape the up-down (U-D direction) height of which is less than the front-rear (F-R direction) length. Additionally, for dry air to spray in a lowermost position, the up-down (U-D direction) height of the discharge opening **833** may remain constant in the front-rear direction (F-R direction).
(206) Additionally, as illustrated, a front edge **833***b* of the discharge opening **833**, which is a first side edge, may be formed on the right side surface of the airflow guide **83**, having a flat plate shape, and a rear edge **833***a* that is a second side edge may extend to the rear surface of the airflow guide **83**, having a curved surface shape. That is, the front edge **833***b* and the rear edge **833***a* of the discharge opening **833** may be spaced from each other, along the direction where an upper end edge of the discharge opening **833** extends.
(207) Accordingly, dry air sprayed through the discharge opening **833** may be discharged in the lowest position with respect to the up-down direction (U-D direction), and based on the slit shape of the discharge opening, spray having directionality, which is approximately parallel with the rear surface **23** of the bottom tub **20***c* with respect to the horizontal direction and does not face the door **30** linearly.

(208) Further, to promote the effect of distributing dry air sprayed through the discharge opening **833**, the discharge opening **833** may extend from the lower portion of the lower rack **51**, between a space S formed between the lower rack **51** and the rear surface **23** of the bottom tub **20***c*.
(209) That is, at least a portion of dry air sprayed through the slit-shaped discharge opening **833** may spray toward between the lower surface **25** of the bottom tub **20***c* and the lower rack **51**, and the remaining portion of the dry air may spray toward the space S.
(210) To this end, the front edge **833***a* of the slit-shaped discharge opening **833** may be disposed under the lower rack **51**, and the rear edge **833***b* of the discharge opening **833** may be disposed in the space S.
(211) In other words, the discharge opening **833** of the airflow guide **83**, as illustrated in FIG. **9**, may be divided into a first portion **8331** disposed under the lower rack **51**, and a second portion **8332** disposed in the space S. and a rear end portion **511** of the lower rack **51** may be a reference line dividing the first portion **8331** and the second portion **8332**.
(212) By doing so, dry air may spray in a direction that does not face the center of the lower rack **51** directly, or a direction that avoids the lower rack **51**, and dry airflow F having passed through the first portion **8331** may move up toward the lower surface of the lower rack **51**, and dry airflow F having passed through the second portion **8332** may move up by passing through the space S.
(213) Thus, the thermal energy of dry air may be distributed evenly in the tub **20** without concentrating on a specific portion of the lower rack **51**.
(214) A relative ratio of the first portion **8331** and the second portion **8332** may be set differently depending on a required distribution ratio of dry air. That is, when more dry air needs to be supplied to the lower rack **51**, a surface area ratio of the first portion **8331** may increase, and when more dry air needs to be supplied to the space S, a surface area ratio of the second portion **8332** may increase.
(215) However, since the capacity of the lower rack **51** is much greater than the capacity of the upper rack **52** or the top rack **53**, the surface area ratio of the first portion **8331** is greater than the surface area ratio of the second portion **8332**, for example.
(216) To this end, the airflow guide **83**, as illustrated in FIGS. **11** and **12**, may be disposed closer to the rear surface **23** of the bottom tub **20***c* than the lower rack **51**. Specifically, when the rear end portion **511** of the lower rack **51** is spaced a first distance G**1** apart from the rear surface **23** of the bottom tub **20***c* in the state where the lower rack **51** is stored in the wash space, the rear end portion of the airflow guide **83** may be spaced from the rear surface **23** of the bottom tub **20***c* to have a second distance G**2** less than the first distance G**1**.
(217) At this time, the front edge **833***a* of the slit-shaped discharge opening **833** may be disposed under the lower rack **51**, and the rear edge **833***b* of the discharge opening **833** may be disposed in the space S.
(218) In other words, the discharge opening **833** of the airflow guide **83**, as illustrated in FIG. **12**, may be divided into a first portion **8331** disposed under the lower rack **51**, and a second portion **8332** disposed in the space S, and the rear end portion **511** of the lower rack **51** may be a reference line dividing the first portion **8331** and the second portion **8332**.
(219) By doing so, dry air may spray in a direction that does not face the center of the lower rack **51** directly, or a direction that avoids the lower rack **51**, and dry airflow F**1** having passed through the first portion **8331** may move up toward the lower surface of the lower rack **51**, and dry airflow F**2** having passed through the second portion **8332** may move up by passing through the space S.
(220) Thus, the thermal energy of dry air may be distributed evenly in the tub **20** without concentrating on a specific portion of the lower rack **51**.
(221) A relative ratio of the first portion **8331** and the second portion **8332** may be set differently depending on a required distribution ratio of dry air. That is, when more dry air needs to be supplied to the lower rack **51**, a surface area ratio of the first portion **8331** may increase, and when more dry air needs to be supplied to the space S, a surface area ratio of the second portion **8332**

may increase.
(222) However, since the capacity of the lower rack **51** is much greater than the capacity of the upper rack **52** or the top rack **53**, the surface area ratio of the first portion **8331** is greater than the surface area ratio of the second portion **8332**, for example.
(223) Considering the fact, the surface area of the second portion **8332**, escaping from the rear end portion **511** of the lower rack **51** and being disposed in the space S, may account for 25% to 50% of the entire surface area of the discharge opening **833** of the airflow guide **83**.
(224) Further, since the rear edge **833***b* of the discharge opening **833**, as illustrated in FIGS. **11** and **12**, extends to the rear surface of the airflow guide **83**, at least a portion of dry airflow sprayed at the rear edge **833***b* side of the discharge opening **833** may have directionality facing the rear surface **23** of the bottom tub **20***c*.
(225) As illustrated in FIG. **13**, the airflow guide **83** may be disposed outside the rotation range R**1** of the lower spray arm **61**, and separated and spaced from the rear surface **23**, the lower surface **25** and the left side surface **26** of the bottom tub **20***c*.
(226) That is, the airflow guide **83** may be disposed between the corner gathered and formed by the rear surface **23**, the lower surface **25** and the left side surface **26** of the bottom tub **20***c*, and the rotation rage of the lower spray arm **61**.
(227) Since the airflow guide **83** is disposed at the corner of the bottom tub **20***c*, outside the rotation range R**1** of the lower spray arm **61**, as described above, interference with the lower spray arm **61** rotating in the washing stage or the rinsing stage may be prevented effectively.
(228) Additionally, since the airflow guide **83** is spaced a predetermined distance apart from the rear surface **23**, the lower surface **25** and the left side surface **26** of the bottom tub **20***c*, food and the like may be effectively prevented from being fitted and fixed between the airflow guide **83** and the bottom tub **20***c*.
(229) FIGS. **14**A to **14**C are views showing temperature distribution measured respectively at a top rack **53**, an upper rack **52** and a lower rack **51** in the state where dry air is supplied through an airflow guide **83** of the related art, and FIGS. **15**A to **15**C are views showing temperature distribution measured respectively at the top rack **53**, the upper rack **52** and the lower rack **51** in the state where dry air is supplied through the airflow guide **83** of one embodiment.
(230) The experiment on the comparative example of the related art in FIGS. **14**A to **14**C, and the experiment on the experimental example of one embodiment in FIGS. **15**A to **15**C were performed under the same test conditions, except for the direction in which dry air sprayed.
(231) As illustrated in FIGS. **14**A to **14**C, when the airflow guide **83** is disposed near the lower surface **25** of the bottom tub **20***c* or dry air sprays toward the central portion of the bottom tub **20***c*, in the related art, there is no big temperature deviation in each position of the lower rack **51** (FIG. **14**A).
(232) However, the temperature deviation in each position of the upper rack **52** and the top rack **53** reveals that temperature decreases rapidly from the central portion of the upper rack **52** and the top rack **53** to the outer side of the upper rack **52** and the top rack **53**, and there is a big deviation between the central portion and the outer side (FIG. **14**B and FIG. **14**C).
(233) On the contrary, as illustrated in FIGS. **15**A to **15**C, when the airflow guide **83** is disposed near the lower surface **25** of the bottom tub **20***c*, but dry air sprays in a direction parallel with the rear surface **23** of the bottom tub **20***c* and is distributed evenly, in the experimental example of the present disclosure, there is no big temperature deviation in each position of the lower rack **51** (FIG. **15**A).
(234) Further, the temperature deviation in each position of the upper rack **52** and the top rack **53** reveals that temperature decreases from the central portion of the upper rack **52** and the top rack **53** to the outer side of the upper rack **52** and the top rack **53** gradually and slowly, and there is no big temperature deviation between the central portion and the outer side.
(235) In particular, unlike the related art, the present disclosure shows that the temperature of the

central portion of the top rack **53** also remains quite high, that the temperature of dry air sprayed along the airflow guide **83** of one embodiment remains constant in the tub **20** entirely/dry air sprayed along the airflow guide **83** of one embodiment may make the temperature inside the entire tub **20** constant and that the effect of drying wash targets is promoted noticeably.
(236) Hereafter, the inner structure of the airflow guide **83** of the dishwasher **1** of one embodiment is described with reference to FIGS. **16** to **24**.
(237) As illustrated in FIG. **16**, the airflow guide **83** of the dishwasher **1** of one embodiment may comprise a lower guide **831** detachably coupled to the duct main body **851** of the connection duct part **85**, an upper guide **832** coupled to the upper side of the lower guide **831**, and a cap cover **834** disposed at the upper side of the upper guide **832** and coupled to the outer surface of the upper guide **832**.
(238) The airflow guide **83**, for example, may be divided with respect to the up-down direction (U-D direction). The lower guide **831** constitutes the lower portion of the segment body. The upper guide **832** and the cap cover **834** may constitute the upper portion of the segment body.
(239) The upper guide **832** is coupled to the upper side of the lower guide **831** described hereafter, and forms a closed inner flow space formed into a channel in which dry airflow F flows together with the lower guide **831**.
(240) To form the inner flow space as illustrated in FIGS. **17** and **18**, the upper guide **832** may be formed into a container which has a vacant space therein, and the lower surface of which is open entirely.
(241) The open lower surface of the upper guide **832** may be coupled with a guide main body **8311** of the lower guide **831** and be closed entirely. By doing so, a closed inner flow space may be formed between the upper guide **832** and the lower guide **831**.
(242) At this time, in response to the shape of the lower guide **831**, the outer shape of the upper guide **832** may have a front-rear width greater than a left-right width.
(243) The upper guide **832** may be formed into a container that has an upper end surface **8321** formed approximately in parallel with a reference surface **8311***a* of the lower guide **831** described hereafter, and a lower surface being open through an outer wall surface which extends in the downward direction (D-direction) along the circumference of the upper end surface **8321**.
(244) At this time, the upper end surface **8321** and the outer wall surface may be integrally formed, and have a uniform thickness as a whole to ensure a maximum inner flow space, and preferably, be manufactured using plastic injection molding.
(245) The outer wall surface may comprise a first flat surface part **8322***c* that forms the right side surface of the outer wall surface and is formed into a flat plate, and a second flat surface part **8322***d* that forms the left side surface of the outer wall surface and is formed into a flat plate. The first flat surface part **8322***c* and the second flat surface part **8322***d* may have a symmetrical shape, and be formed into a perpendicular surface or an inclined surface having a slope where a gap between the first flat surface part **8322***c* and the second flat surface part **8322***d* decreases gradually in the upward direction (U-direction).
(246) Additionally, a first curved surface part **8322***a* may be continuously formed at the front sides of the first flat surface part **8322***c* and the second flat surface part **8322***d*, and a second curved surface part **8322***b* may be continuously formed at the rear sides of the first flat surface part **8322***c* and the second flat surface part **8322***d*.
(247) The first curved surface part **8322***a* may form the front surface of the upper guide **832**, and for example, have an outer shape of a half cylindrical surface that is convex forward.
(248) Like the first curved surface part **8322***a*, the second curved surface part **8322***b* may form the rear surface of the upper guide **832**, and for example, have an outer shape of a half cylindrical surface that is convex rearward.
(249) The first curved surface part **8322***a* and the second curved surface part **8322***b* may be disposed to have an approximately symmetrical shape.

(250) Each of the first curved surface part **8322***a* and the second curved surface part **8322***b* may integrally connect to the upper end surface **8321**, the first flat surface part **8322***c* and the second flat surface part **8322***dc*, and form a continuous surface for the upper end surface **8321**, the first flat surface part **8322***c* and the second flat surface part **8322***d*.
(251) As illustrated, a first camper surface **8322***e* in a camper shape may be formed at an edge formed between the upper end surface **8321** and the outer wall surface. The first camper surface **8322***e* may help to minimize flow loss or noise caused by eddy currents that may be generated at an angular edge side of the inner flow space in which dry airflow F flows.
(252) The first camper surface **8322***e* may be a curved surface having a predetermined curvature, or an inclined surface having a predetermined slope.
(253) Like the first camper surface **8322***e*, a second camper surface **8322***f* in a camper shape may be formed at an edge that is formed by the upper end surface **8321** and the first curved surface part **8322***a* which are met, and at an edge that is formed by the upper end surface **8321** and the second curved surface part **8322***b* which are met.
(254) Like the first camper surface **8322***e*, the second camper surface **8322***f* may be a curved surface having a predetermined curvature, or an inclined surface having a predetermined slope.
(255) A lower end portion **8323** of the upper guide **832** formed around the open lower surface of the upper guide **832** may be formed continuously while its height remains approximately constant from the upper end surface **8321** with respect to the up-down direction (U-D direction) such that the lower end portion **8323** of the upper guide **832** may be inserted and coupled to a misassembly prevention groove **8311***d* of the lower guide **831** described hereafter.
(256) A first notch hole **8324** forming the front edge, the rear edge and the upper end edge of the discharge opening **833** may be formed in the lower end portion **8323** of the upper guide **832**.
(257) The first notch hole **8324**, as illustrated, may be formed into a notch where the first flat surface part **8322***c* and the second curved surface part **8322***b* of the upper guide **832** are partially cut.
(258) The lower end of the first notch hole **8324** is entirely open, and as the upper guide **832** is coupled to the lower guide **831**, the above-described reference surface **8311***a* of the lower guide **831** may be coupled to the open lower end of the first notch hole **8324**, and the reference surface **8311***a* may block the lower end of the first notch hole **8324**.
(259) The upper end edge of the first notch hole **8324** may extend approximately in parallel with the reference surface **8311***a* of the lower guide **831** and may extend linearly.
(260) The front edge of the first notch hole **8324** may be formed at the first flat surface part **8322***c* and extend linearly along the up-down direction (U-D direction). The rear edge of the first notch hole **8324** may be formed at the second curved surface part **8322***b* and extend linearly along the up-down direction (U-D direction).
(261) A rear corner part formed by the upper end edge and the rear edge which are met, and a front corner part formed by the upper end edge and the front edge which are met may respectively have a curved edge having a predetermined curvature.
(262) A first holding hole **8325** which is formed into a rectangular penetration hole and to which the upper guide holding projection **8312***g* of the lower guide **831** is held and coupled may be formed at the first curved surface part **8322***a* of the upper guide **832**, near the lower end portion **8323** of the upper guide **832**.
(263) As illustrated in FIG. **17**, the outer shape of the upper guide **832** is approximately symmetrical with respect to the front-rear direction (F-R direction), but the first notch hole **8324** and the first holding hole **8325** are formed in an asymmetrical position with respect to the front-rear direction (F-R direction). The first notch hole **8324** and the first holding hole **8325** may serve as a means of preventing the misassembly of the upper guide **832** to the lower guide **831**.
(264) Additionally, at least one cap cover holding projection **8326** for fastening the cap cover **834** described hereafter may be integrally formed at the second flat surface part **8322***d* of the upper

guide **832**.
(265) As described hereafter, the cap cover **834** may be coupled to the outer surface of the upper guide **832**. At least one cap cover holding projection **8326** may have a lamp surface having a predetermined angle with respect to the second flat surface part **8322***d*, and a step surface formed approximately perpendicularly with respect to the second flat surface part, to ensure ease of coupling and prevent ease of separation.
(266) As illustrated in FIG. **16**, the cap cover **834** may be provided with a second holding hole **8345** that is formed near the lower end portion **8343** in a way that penetrates the inside and the outside of the cap cover **834**.
(267) The second holding hole **8345** may be formed into a rectangular penetration hole having a width and a height at which the cap cover holding projection **8326** can be inserted into the second holding hole **8345** at a time of coupling the upper guide **832** and the cap cover **834**, and like the cap cover holding projection **8326**, may be disposed higher than the upper guide holding projection **8312***g* with respect to the up-down direction.
(268) By doing so, the cap cover holding projection **8326** and the second holding hole **8345** may clearly distinguish from the upper guide holding projection **8312***g* because of the difference in their heights, and the misassembly of the cap cover **834** having a symmetrical shape with respect to the front-rear direction (F-R direction) may be prevented effectively.
(269) Additionally, the upper guide **832** of the airflow guide **83** of one embodiment may be provided with a blocking rib **8328** as a first means of minimizing the inflow of wash water into the airflow guide **83** and the connection duct part **85** through the discharge opening **833**.
(270) As illustrated in FIGS. **17** and **18**, the blocking rib **8328** may comprise a first rib **8328***a* that extends in a shade shape along the upper end edge, the front edge and the rear edge of the first notch hole **8324**.
(271) The first rib **8328***a* extends continuously along the edges of the first notch hole **8324** and protrude approximately perpendicularly with respect to the first flat surface part **8322***c* of the upper guide **832**. Preferably, the first rib **8328***a* may be formed integrally on the first flat surface part **8322***c* and the second curved surface part **8322***b* of the upper guide **832**.
(272) One end portion of the first rib **8328***a* may be formed at the rear edge of the first notch hole **8324**, and the other end portion of the first rib **8328***a* may be formed at the front edge of the first notch hole **8324**. The first rib **8328***a* may extend in a continuous protruding wall shape, between one end portion and the other end portion thereof, to serve as a shade surrounding the first notch hole **8324** approximately entirely.
(273) However, to prevent interference with a first edge wall **8311***b* of the lower guide **831**, one end portion and the other end portion of the first rib **8328***a* may be respectively spaced a predetermined height from the lower end portion **8323** of the upper guide **832** in the upward direction (U-direction).
(274) The first rib **8328***a* may help to minimize the fall of wash targets from the storage part after a wash or the passage of wash water scattered after a fall through the first notch hole **8324**.
(275) To this end, a horizontal rib of the first rib **8328***a*, formed at least at the upper end edge of the first notch hole **8324**, may horizontally protrude past the first edge wall **8311***b* of the lower guide **831** and the reference surface **8311***a*, as illustrated in FIG. **17**.
(276) That is, the horizontal portion of the first rib **8328***a* may extend to cover the first edge wall **8311***b* and the reference surface **8311***a* with respect to the up-down direction (U-D direction), and in the state where the airflow guide **83** is installed completely, the first edge wall **8311***b* and the reference surface **8311***a* are covered and by the first rib **8328***a* and is not be seen, when view from above.
(277) Accordingly, the flow of wash water, which falls perpendicularly after wash targets are washed, collides with the first edge wall **8311***b* and then is scattered, into the first notch hole **8324** may be minimized.

(278) However, wash water scattered in the washing stage or the rinsing stage may fall in a direction different from the perpendicular direction. That is, wash water avoiding the first rib **8328***a*, colliding with the first edge wall **8311***b* and being scattered is likely to flow into the first notch hole **8324**.
(279) To prevent this from happening, the blocking rib **8328** may further comprise at least one second rib **8328***b* that extends across the inside the first notch hole **8324**, along the front-rear direction (F-R direction).
(280) FIGS. **17** to **18** exemplarily show an embodiment provided with a pair of second ribs **8328***b* that are spaced in the up-down direction (U-D direction). Hereafter, an embodiment provided with a pair of second ribs **8328***b*, as illustrated, is described for convenience, but not limited.
(281) Each of the pair of second ribs **8328***b* may extend across the inside the first notch hole **8324** and have the same shape.
(282) At this time, to prevent deterioration in the spray efficiency of dry air spraying, the up-down thickness of each of the second ribs **8328***b* may be much less than the front-rear length.
(283) Additionally, to minimize the flow of perpendicularly falling wash water into the first notch hole **8324**, caused by the collision with the first edge wall **8311***b* and scattering of the wash water, the pair of second ribs **8328***b*, like the first rib **8328***a*, may protrude horizontally past the first edge wall **8311***b* and the reference surface **8311***a* of the lower guide **831**. That is, the horizontal position of the left end portion of the horizontal rib of the first rib **8328***a* may be the same as the horizontal position of the left end portion of the pair of second ribs **8328***b*.
(284) However, to cover the inside of the first notch hole **8324** entirely with respect to the front-rear direction (F-R direction), the rear end portion forming one end portion of each of the second ribs **8328***b* may integrally connect to the rear edge of the first notch hole **8324**, and the front end portion forming the other end portion of each of the second rib **8328***b* may integrally connect to the front edge of the first notch hole **8324**.
(285) The shape of the second rib **8328***b* may help to prevent the inflow of wash water, in a way that the wash water is blocked by the second rib **8328***b* again, even if the wash water collides with the first edge wall **8311***b* by avoiding the first rib **8328***a* and is scattered.
(286) Additionally, since the thickness of the pair of second ribs **8328***b* is much less than the front-rear length thereof, the second ribs have relatively low strength, and is likely to be damaged by small magnitude of external force. To prevent such damage, a bridge rib **8328***c* is disposed between the front end portion and the rear end portion of the second rib **8328***b* and connects the pair of second ribs **8328***b* mutually to reinforce the second ribs. In the illustrative embodiment, the bridge rib **8328***c* extends only between the pair of second ribs **8328***b*, but may further extend to the upper end edge of the first notch hole **8324**.
(287) Further, the upper guide **832** of the airflow guide **83** of one embodiment may be provided with at least one blocking wall **8329** that is disposed in the inner flow space of the upper guide **832**, as a second means of minimizing the flow of wash water into the airflow guide **83** and the connection duct part **85** through the discharge opening **833**.
(288) At least one blocking wall **8329** may help to prevent and minimize the movement of the droplets of wash water, which has passed through the blocking rib **8328** and flown into the airflow guide **83** after the wash water's collision and scattering, toward the lower portion of the upper end surface **8321** of the upper guide **832** or toward a duct coupling part **8312** of the lower guide **831**.
(289) To this end, as illustrated in FIGS. **17** and **18**, at least one blocking wall **8329** may be disposed in the form of a barrier that extends downward from the lower portion of the upper end surface **8321** of the upper guide **832**, to at least partially block the upper portion side of the first notch hole **8324** forming the discharge opening **833**.
(290) That is, the up-down position of the lower end of at least one blocking wall **8329** may be between the upper end edge of the first notch hole **8324** and the reference surface **8311***a* of the lower guide **831**. Accordingly, when the inside of the first notch hole **8324** is viewed from the

outside, the duct coupling part **8312** of the lower guide **831** is entirely covered by at least one blocking wall **8329** and is not be seen visually from the outside.

(291) Thus, among droplets of wash water being scattered and flowing into the airflow guide **83**, droplets bouncing upward collide with at least one blocking wall **8329** and flow downward along at least one blocking wall **8329** by using gravity.

(292) The droplets prevented from coming in and falling downward along at least one blocking wall **8329** need to be discharged out of the airflow guide **83** again. To this end, the lower end edge of at least one blocking wall **8329** may extend toward a channel guide surface **8313** of the lower guide **831**. Additionally, at least one blocking wall **8329** may be disposed in the channel guide surface area with respect to the horizontal direction, as described hereafter. A relative position relationship between the blocking wall **8329** and the channel guide surface **8313** of the lower guide **831** is described hereafter with reference to FIGS. **23** and **24**.

(293) Further, as shown in the illustrative embodiment, the blocking wall **8329** may comprise a first blocking wall **8329***a* disposed relatively close to the first notch hole **8324**, and a second blocking wall **8329***b* disposed farther from the first notch hole **8324** than the first blocking wall **8329***a*, for example.

(294) As described above, at least one blocking wall **8329** is disposed in the inner flow space where dry airflow F flows.

(295) Accordingly, if a single blocking wall covers the upper portion side of the first notch hole **8324** entirely, the flow resistance of dry airflow F may increase, resulting in deterioration of air blowing efficiency.

(296) To prevent deterioration in air blowing efficiency, caused by an increase in flow resistance, the blocking wall **8329** may be divided into the first blocking wall **8329***a* and the second blocking wall **8329***b* to complementarily cover the upper portion side of the first notch hole **8324**.

(297) That is, the first blocking wall **8329***a* may be disposed to partially cover the upper portion side of the first notch hole **8324**, and the second blocking wall **8329***b* may be disposed to at least partially cover the remaining portion of the first notch hole **8324** that is not cover by the first blocking wall **8329***a*.

(298) The upper end edge of the first blocking wall **8329***a* may integrally connect to the lower portion of the upper end surface **8321** of the upper guide **832** and be formed into a curved surface that is convex in a direction farther from the first notch hole **8324** forming the discharge opening **833**, from one end edge **8329***a***1** of the first blocking wall **8329***a* toward the other end edge **8329***a***2** thereof.

(299) Likewise, the upper end edge of the second blocking wall **8329***b* may integrally connect to the lower portion of the upper end surface **8321** of the upper guide **832** and be formed into a curved surface that is convex in a direction farther from the first notch hole **8324** forming the discharge opening **833**, from one end edge **8329***b***1** of the second blocking wall **8329***b* toward the other end edge **8329***b***2** thereof.

(300) That is, the flow direction of air having passed through the upper end **8511** of the connection duct part **85** and flown into the inner flow space changes toward the first notch hole **8324** forming the discharge opening **833**, as described hereafter. That is, in the process of changing a flow direction, a rotation speed component is produced in dry airflow F.

(301) To prevent a rapid change in the flow space and produce a rotation speed component effectively in the process of changing a flow direction, the first blocking wall **8329***a* and the second blocking wall **8329***b* may also serve as a flow guide.

(302) To minimize flow resistance and serve as a flow guide effectively, the first blocking wall **8329***a* and the second blocking wall **8329***b* may be formed into a curved surface having predetermined curvature, and have a uniform thickness from one end edge **8329***a***1**, **8329***b***1** to the other end edge **8329***a***2**, **8329***b***2** of each of the first blocking wall **8329***a* and the second blocking wall **8329***b*.

(303) However, detailed shapes of the first blocking wall **8329***a* and the second blocking wall may differ from each other.
(304) That is, one end edge **8329***a***1** of the first blocking wall **8329***a* may be a fixation end portion that integrally connects to the front edge of the first notch hole **8324**, and the other end edge **8329***a***2** of the first blocking wall **8329***a* may be a free end portion that does not connect to the outer wall surface of the upper guide **832**.
(305) One end edge **8329***a***1** of the first blocking wall **8329***a* integrally connects to the front edge of the first notch hole **8324**, as described above, such that droplets of wash water are prevented from flowing directly into an upper end **8312***a* of the duct coupling part **8312** of the lower guide **831** and the upper end **8511** of the connection duct part **85** through the front edge side of the first notch hole **8324**, in a minimum distance.
(306) Additionally, the up-down position of a lower end edge **8329***a***3** of the first blocking wall **8329***a* may change from one end edge **8329***a***1** of the first blocking wall **8329***a* to the other end edge **8329***a***2** thereof. For example, the lower end edge **8329***a***3** of the first blocking wall **8329***a* may have a step.
(307) Specifically, the lower end edge **8329***a***3** of the first blocking wall **8329***a* may comprise a first edge **8329***a***31** the up-down position of which is maintained approximately in a first position, and a second edge **8329***a***32** the up-down position of which is maintained approximately in a second position. At this time, the first position is lower than the second position, thereby forming a step.
(308) The first edge **8329***a***31**, as illustrated, may be disposed closer to the one end edge of the first blocking wall **8329***a* and the front edge side of the first notch hole **8324** than the second edge **8329***a***32**.
(309) That is, the up-down position of the lower end edge of the first blocking wall **8329***a*, formed near the front edge side of the first notch hole **8324**, may remain lower. By doing so, droplets of wash water may be further prevented from directly flowing into the upper end **8312***a* of the duct coupling part **8312** of the lower guide **831** and the upper end **8511** of the connection duct part **85**, into which dry airflow F comes through the front edge side of the first notch hole **8324**, in a minimum distance.
(310) As described hereafter, the first position of the first edge **8329***a***31** and the second position of the second edge **8329***a***32** may be lower than the up-down position of the upper end of a division wall provided at the lower guide **831**. That is, the lower end edge **8329***a***3** of the first blocking wall **8329***a* may entirely extend to a position lower than the up-down position of the upper end of the division wall **8314**.
(311) Additionally, the second blocking wall **8329***b* may be disposed in the state of separating from the first blocking wall **8329***a* and disposed between the first blocking wall **8329***a* and the second flat surface part **8322***d*.
(312) As illustrated, each of one end edge **8329***b***1** and the other end edge **8329***b***2** of the second blocking wall **8329***b* may be a free end portion that does not connect to the outer wall surface of the upper guide **832**.
(313) However, the other end edge **8329***b***2** of the second blocking wall **8329***b* may be disposed closer to the rear edge of the first notch hole **8324** than the other end edge **8329***a***2** of the first blocking wall **8329***a* with respect to the front-rear direction, as described above.
(314) Further, the up-down position of a lower end edge **8329***b***3** of the second blocking wall **8329***b* may remain constant approximately from one end edge **8329***b***1** to the other end edge **8329***b***2** and remain higher than the second edge **8329***a***32** of the first blocking wall **8329***a*.
(315) However, like the up-down position of the lower end edge of the first blocking wall **8329***a*, the up-down position of the lower end edge **8329***b***3** of the second blocking wall **8329***b* may be entirely lower than the up-down position of the upper end of the division wall **8314** provided at the lower guide **831**.
(316) Further, the cap cover **834** disposed at the upper side of the upper guide **832** may be coupled

to the outer surface of the upper guide **832** to protect the upper guide **832**.
(317) The upper guide **832** is disposed lower than the storage part that accommodates wash targets, and the upper end surface **8321** and the outer wall surface are disposed in a way that the upper end surface **8321** and the outer wall surface are mostly exposed to the wash space **21** of the tub **20**. However, as described above, the upper guide **832** is made of a plastic material having relatively low strength.
(318) Accordingly, the upper guide **832** may be broken directly due to a collision with wash targets that may fall from the storage part between the washing stage and the rinsing stage or may fall while the user withdraws the storage part.
(319) The cap cover **834** is coupled to the upper side of the outer surface of the upper guide **832** to prevent the damage to the upper guide **832**, caused by a collision with wash targets.
(320) To this end, the cap cover **834** may be made of a material having higher breaking strength and corrosion resistance than the upper guide **832**, and preferably, may be formed with a sheet of metal such as stainless steel and the like.
(321) To be coupled to the outer surface of the upper guide **832**, the cap cover **834** may have a shape corresponding to the shape of the outer surface of the upper guide **832**.
(322) Accordingly, like the upper guide **832**, the cap cover **834** has a vacant space therein, and is formed into a container that is entirely open.
(323) The upper guide **832** may be inserted and coupled through an open lower surface of the cap cover **834**.
(324) In response to the shape of the upper guide **832**, the outer shape of the cap cover **834** may have a front-rear width greater than a left-right width.
(325) Specifically, the cap cover **834** may comprise an upper end surface **8341** formed in parallel with the upper end surface **8321** of the upper guide **832**, and an outer wall surface extending along the circumference of the upper end surface **8321** in the downward direction (D-direction).
(326) Like the upper guide **832**, the outer wall surface of the cap cover **834** may comprise a first flat surface part **8342***c* forming a right side surface and being formed into a flat plate, and a second flat surface part **8342***d* forming a left side surface and being formed into a flat plate.
(327) The first flat surface part **8342***c* and the second flat surface part **8342***d* may have a symmetrical shape, and be formed into a perpendicular surface or an inclined surface having a slope where a gap between the first flat surface part **8342***c* and the second flat surface part **8342***d* decreases gradually in the upward direction (U-direction).
(328) Additionally, a first curved surface part **8342***a* may be formed at the front sides of the first flat surface part **8342***c* and the second flat surface part **8342***d*, and a second curved surface part **8342***b* may be continuously formed at the rear sides of the first flat surface part **8342***c* and the second flat surface part **8342***d*.
(329) The first curved surface part **8342***a* may form the front surface of the cap cover **834**, and for example, have an outer shape of a half cylindrical surface that is convex forward in response to the shape of the first curved surface part **8322***a* of the upper guide **832**.
(330) The second curved surface part **8342***b* may form the rear surface of the cap cover **834**, and for example, have an outer shape of a half cylindrical surface that is convex forward in response to the shape of the second curved surface part **8322***b* of the upper guide **832**.
(331) Each of the first curved surface part **8342***a* and the second curved surface part **8342***b* may be integrally formed on the upper end surface **8341**, the first flat surface part **8342***c* and the second flat surface part **8342***d*, and form a continuous surface for the upper end surface **8341**, the first flat surface part **8342***c* and the second flat surface part **8342***d*.
(332) Additionally, in response to the upper guide **832**, a camper surface **8342***e* in a camper shape may be formed at an edge formed between the upper end surface **8341** and the outer wall surface.
(333) However, unlike the upper guide **832**, the cap cover **834** is not provided with a component corresponding to the second camper surface **8322***f* of the upper guide **832**.

(334) The lower end portion **8343** of the outer wall surface of the cap cover **834** may extend to the lower end portion **8323** of the outer wall surface of the upper guide **832** to cover the outer wall surface of the upper guide **832** entirely. Accordingly, at a time of coupling the cap cover **834** to the lower guide **831**, the lower end portion **8343** of the cap cover **834** and the lower end portion **8323** of the upper guide **832** may be inserted into the misassembly prevention groove **8311***d* of the lower guide **831**, together.
(335) Additionally, a second notch hole **8344** may be formed in the lower end portion **8343** of the cap cover **834** and have a shape corresponding to that of the first notch hole **8324** of the upper guide **832**.
(336) Like the first notch hole **8324**, the second notch hole **8344** may be formed into a notch where the first flat surface part **8342***c* and the second curved surface part **8342***b* of the cap cover **834** are partially cut.
(337) Since the second notch hole **8344** has the same shape as the first notch hole **8324**, a detailed shape of the second notch hole **8344** is not described.
(338) However, a holding jaw **8344***a* may be provided at the lower end of the front edge of the second notch hole **8344** and protrude to the inside of the second notch hole **8344**.
(339) The holding jaw **8344***a* is a portion that is held and coupled to the other end portion of the above-described first rib **8328***a*, and the right side portion of the cap cover **834** may be coupled to the upper guide **832** through the holding jaw **8344***a*.
(340) Additionally, the second holding hole **8345** may be formed near the lower end portion **8323** of the upper guide **832**, at the second flat surface part **8342***d* of the cap cover **834**, and formed into a rectangular penetration hole to which the above-described cap cover holding projection **8326** of the upper guide **832** is held and coupled.
(341) As the cap cover holding projection **8326** is held and coupled to the second holding hole **8345**, the left side portion of the cap cover **834** may be coupled to the upper guide **832**.
(342) That is, the cap cover **834** may be coupled to two spots of the upper guide **832** at least though the holding jaw **8344***a* and the cap cover holding projection **8326**.
(343) Further, the lower guide **831**, as illustrated in FIGS. **19** to **21**, may comprise a guide main body **8311** formed into an approximately flat plate.
(344) The guide main body **8311** may have an outer shape in which a front-rear (F-R direction) width is greater than a left-right (Le-Ri direction) width, in the state where the guide main body **8311** is disposed at the connection duct part **85**.
(345) At this time, the left and right edges of the outer edge forming the outer shape of the guide main body **8311** may have a linear shape, the front edge may have a circular arc shape that is convex forward, and the rear edge may have a circular arc shape that is convex rearward.
(346) The left and right edges of the guide main body **8311** may have shapes that are approximately symmetrical to each other and parallel with each other, and the front and rear edges of the guide main body **8311** may have shapes that are symmetrical to each other.
(347) The reference surface **8311***a* serving as a lower end edge of the above-described discharge opening **833** may be formed at the right edge side of the guide main body **8311**. The reference surface **8311***a*, as illustrated, may be provided in the form of a flat surface that extends in a direction farther from the discharge opening **833** along the horizontal direction, and extend to the lower end of the channel guide surface **8313** described hereafter from the right edge.
(348) Further, the first edge wall **8311***b* may be formed at least partially at the left, right, front and rear edges of the guide main body **8311** and extend from the reference surface **8311***a* in the upward direction (U-direction) by a predetermined height.
(349) The first edge wall **8311***b*, as illustrated, may be formed continuously along the outer edge of the guide main body **8311**. However, the first edge wall **8311***b* may not be formed at least in the discharge opening area **833** not to prevent the spray of dry air, as illustrated in FIG. **19**.
(350) Further, a misassembly prevention groove **8311***d* may be formed inside the first edge wall

8311_b_ and be depressed further downward (in the D-direction) than the reference surface **8311**_a_, with respect to the up-down direction (U-D direction), and serve as a misassembly prevention part of the upper guide **832**. At this time, the height at which the misassembly prevention groove **8311**_d_ is depressed from the reference surface **8311**_a_, may remain constant approximately along the first edge wall **8311**_b_.

(351) The lower end portion **8323** of the upper guide **832**, which is described hereafter, may be inserted and coupled to the misassembly prevention groove **8311**_d_. Accordingly, the misassembly prevention groove **8311**_d_ may have a shape and a size corresponding to the shape and the size of the lower end portion **8323** of the upper guide **832**. As described above, the shape of the lower end portion **8323** of the upper guide **832** is formed continuously except for the area where the first notch hole **8324** forming the discharge opening **833** is formed, i.e., the area where the reference surface **8311**_a_ is formed. In response, the misassembly prevention groove **8311**_d_ may be formed continuously along the first edge wall **8311**_b_.

(352) At this time, the shape of the lower end portion **8323** of the upper guide **832** may be asymmetrical to the shape of the misassembly prevention groove **8311**_d_ with respect to the front-rear direction (F-R direction). Accordingly, the lower end portion **8323** of the upper guide **832** may not be coupled and fastened to the misassembly prevention groove **8311**_d_ in a direction different from a predetermined direction. By doing so, a misassembly between the upper guide **832** and the lower guide **831** may be prevented effectively.

(353) Further, a second edge wall **8311**_c_ may be formed at the front edge of the guide main body **8311** and extend from the reference surface **8311**_a_ in the downward direction (D-direction) to have a predetermined height.

(354) The second edge wall **8311**_c_, as illustrated, may be formed continuously into a cylinder along the circular arc-shaped front edge of the guide main body **8311**, and the lower end potion of the second edge wall **8311**_c_ may extend past a lower end **8312**_b_ of the duct coupling part **8312** that is described below.

(355) That is, the second edge wall **8311**_c_ may be formed in a way that surrounds the outer surface of the below-described duct coupling part **8312** at least partially. At this time, the second edge wall **8311**_c_ is formed in the state of being separated and spaced from the duct coupling part **8312**, and a predetermined space may be formed between the second edge wall **8311**_c_ and the duct coupling part **8312**. As described below, an upper end **8522** of a fastening nut **852** may be inserted into the space at least partially.

(356) A release prevention part **8311**_e_ may be provided on the second edge wall **8311**_c_, and based on an interaction with the fastening nut **852**, keep the lower guide **831** fixed to the fastening nut **852** and prevent the lower guide **831** from escaping from a fixed position.

(357) As described hereafter, the lower guide **831** is detachably coupled to the duct main body **851** of the connection duct part **85** based on a two-stage coupling manipulation, without an additional coupling member. The two-stage coupling manipulation may comprise an up-down perpendicular movement manipulation and a circumferential rotational movement manipulation, for example.

(358) The release prevention part **8311**_e_ prevents a relative rotation of the lower guide **831** in a direction opposite to the direction of the rotational movement in the two-stage coupling manipulation, i.e., prevents the lower guide **831** from escaping from the fixed position after the second-stage coupling manipulation including the perpendicular movement manipulation and the rotational movement manipulation is completed.

(359) The release prevention part **8311**_e_, as illustrated exemplarily, may be formed integrally on the second edge wall **8311**_c_, and prevent the relative rotation of the lower guide **831** in a direction opposite to the direction of the rotational movement of the two-stage coupling manipulation, in the form of an elastic hook.

(360) The lower guide **831**, as described above, is directly coupled to the duct main body **851** of the connection duct part **85**, using a pipe coupling method.

(361) To this end, the lower guide **831** may comprise a cylindrical duct coupling part **8312** to which the upper end **8511** of the cylinder-shaped duct main body **851** is inserted and detachably coupled.
(362) In response to the shape of the duct main body **851**, the duct coupling part **8312** may be formed into a cylinder the central axis C of which extends in parallel with the up-down direction (U-D direction). For the upper end **8511** of the duct main body **851** to be inserted into and pass through the duct coupling part **8312**, the inner diameter of the lower end **8312***b* of the duct coupling part **8312** may be greater than or the same as the outer diameter of the upper end **8511** of the duct main body **851**.
(363) The duct coupling part **8312** may be formed integrally at the guide main body **8311**, and disposed near the circular arc-shaped rear edge of the guide main body **8311**. That is, the duct coupling part **8312** may be biased toward the rear side of the guide main body **8311** with respect to the front-rear direction (F-R direction).
(364) The upper end **8312***a* of the cylindrical duct coupling part **8312**, from which dry air is discharged, may be formed in a position that protrudes from the guide main body **8311**, in the upward direction (U-direction). Preferably, the upper end **8312***a* of the duct coupling part **8312** protrudes to and is exposed to the inner flow space formed between the guide main body **8311** of the lower guide **831** and the upper guide **832**.
(365) At this time, in the state where the coupling of the upper end **8511** of the connection duct part **85** is completed, the position of the upper end **8312***a* of the duct coupling part **8312** may be formed in a position lower than the position of the upper end **8511** of the connection duct part **85** with respect to the up-down direction (U-D direction).
(366) That is, the upper end **8511** of the connection duct part **85**, as illustrated in FIG. **22**, may protrude further upward than the upper end **8312***a* of the duct coupling part **8312**. Since the upper end **8511** of the connection duct part **85** remains higher than the upper end **8312***a* of the duct coupling part **8312** as described above, the upward movement of droplets of wash water coming in through the discharge opening **833** along a gap between the outer circumferential surface of the connection duct part **85** and the inner circumferential surface of the duct coupling part **8312**, caused by a capillary phenomenon, may be fundamentally blocked.
(367) Additionally, the central axis of the duct coupling part **8712** may be spaced from the first notch hole **8724** with respect to the front-rear direction (F-R direction) or the horizontal direction. In the embodiment, the central axis of the duct coupling part **8712** may be disposed further forward than the first notch hole **8724**. That is, at least the front edge of the first notch hole **8724** is disposed further rearward than the central axis of the duct coupling part **8712**. Accordingly, a portion of the upper end **8712***a* of the duct coupling part **8712**, exposed outward through the discharge opening **873** or the first notch hole **8724**, may be minimized, and the flow of the reversely incoming wash water, having passed through the discharge opening **873**, into the duct coupling part **8712** through the upper end **8712***a* of the duct coupling part **8712** may be minimized.
(368) Further, a first guide groove **8312***d* that extends in a linear shape along the up-down direction (U-D direction), and a second guide groove **8312***e* that extends in a circular arc shape along the circumferential direction may be formed on an inner circumferential surface **8312***c* of the duct coupling part **8312**.
(369) As illustrated, the upper end of the first guide groove **8312***d* integrally connects to one end portion of the second guide groove **8312***e*.
(370) The lower guide **831**, as described above, is coupled to the duct main body **851** of the connection duct part **85**, based on the two-stage coupling manipulation comprising the up-down perpendicular movement manipulation and the circumferential rotational movement manipulation.
(371) The first guide groove **8312***d* extending along the up-down direction (U-D direction) in a linear shape guides the up-down perpendicular movement of the lower guide **831**, and the second guide groove **8312***e* extending along the circumferential direction in a circular arc shape guides the circumferential rotational movement of the lower guide **831**.

(372) A guide projection **8516** may be integrally provided on the outer circumferential surface of the duct main body **851** inserted into and coupled to the duct coupling part **8312** of the lower guide **831**, and protrude toward the inner circumferential surface of the connection duct part **85** and be inserted into the first guide groove **8312*d*** and the second guide groove **8312*e*** of the connection duct part **85**.
(373) Accordingly, the guide projection **8516**, as described hereafter, may be first inserted into the first guide groove **8312*d*** at a time of coupling the lower guide **831** and the duct main body **851**.
(374) As a result, in the state where the guide projection **8516** of the duct main body **851** is inserted into the first guide groove **8312*d***, the lower guide **831** may move perpendicularly in the downward direction (D-direction).
(375) As the up-down perpendicular movement manipulation of the two-stage coupling manipulation starts, the first guide groove **8312*d*** moves in the downward direction (D-direction) along the guide projection **8516** that stands still. As the guide projection **8516** reaches the upper end of the first guide groove **8312*d***, the lower guide **831** may not move in the downward direction (D-direction) any longer because of the guide projection **8516**'s action.
(376) At this time, since the guide projection **8516** has arrived at one end portion of the second guide groove **8312*e***, the lower guide **831** may not make a downward (D-direction) movement while making a circumferential rotational movement in the two-stage coupling manipulation.
(377) As the lower guide **831** rotates for the circumferential rotational movement manipulation of the two-stage coupling manipulation, the second guide groove **8312*e*** moves along the guide projection **8516** that stands still. As the guide projection **8516** reaches the other end portion of the second guide groove **8312*e***, the lower guide **831** may not rotate in the circumferential direction any longer because of the guide projection **8516**'s action.
(378) When the lower guide **831** does not rotate any longer as described above, the lower guide **831** and the duct main body **851** may be coupled completely, the lower guide **831** may be disposed in a fixed position completely, and without an additional coupling member or an additional fastening member, the lower guide **831** and the duct main body **851** may be coupled.
(379) As illustrated in FIGS. **19** and **20**, a stopper projection **8312*f*** may be integrally formed on the inner circumferential surface **8312*c*** of the duct coupling part **8312**, at the other end portion side of the second guide groove **8312*e***, and may form stick-slip in relation to the movement of the guide projection **8516**, and after the guide projection **8516** reaches the other end portion of the second guide groove **8312*e***, stop a relative rotation of the lower guide **831** in the opposite direction.
(380) Thus, as long as external force of greater than a specific level is not applied, the stopper projection **8312*f*** may prevent the guide projection **8516** from escaping from the other end portion of the second guide groove **8312*e***.
(381) However, when the coupling of the lower guide **831** and the setting of the right position of the lower guide **831** are completed in the state where the guide projection **8516** is inserted into the second guide groove **8312*e***, the airflow guide **83** may not make a self weight-induced movement due to a hold between the second guide groove **8312*e*** and the guide projection **8516**.
(382) However, as strong external force such as a collision of a wash target falling in the washing stage or the rinsing stage and the like is applied, the second guide groove **8312*e*** and the guide projection **8516** may be easily released from the hold therebetween.
(383) Even without strong external force, the lower guide **831** and the airflow guide **83** are highly likely to clatter because of a gap between the duct coupling part **8312** of the lower guide **831** and the duct main body **851** of the connection duct part **85**, caused by manufacturing tolerance.
(384) The clatter occurs due to a gap-induced relative displacement or relative movement of the lower guide **831** with respect to the connection duct part **85**.
(385) According to the present disclosure, at least one protruding rib may be included as a means of minimizing a gap between the lower guide **831** and the connection duct part **85**, in particular, a means of limiting a relative downward movement to the connection duct part **85**.

(386) Referring to FIG. **21**, at least one protruding rib may protrude downward from the lower end **8312***b* of the duct coupling part **8312** toward a male screw thread **8541** provided on the outer circumferential surface of the duct main body **851** of the connection duct part **85**.
(387) That is, the airflow guide **83** of the dishwasher of one embodiment may adjust the amount of a generated gap and limit a relative downward movement through the male screw thread **8514** of the duct main body **851** disposed at the lower portion side of the duct coupling part **8312**.
(388) Accordingly, as a relative downward movement of the airflow guide **83**, i.e., a relative downward movement of the lower guide **831**, is made, the relative movement of the lower guide **831** may be limited in a way that the lower end surface of at least one protruding rib contacts one side surface of the male screw thread **8514**.
(389) FIG. **21** shows at least one protruding rib comprising a first protruding rib **8312***h***1**, a second protruding rib **8312***h***2** and a third protruding rib **8312***h***3** that are disposed around a circular opening formed at the lower end **8312***b* of the duct coupling part **8312** of the lower guide **831** at regular intervals, for example. Hereafter, at least one protruding rib comprising the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** is described but not limited.
(390) As illustrated, the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3**, disposed at regular intervals along the circumferential direction, may have the same cross-sectional shape. For example, a cross section in a direction perpendicular to the direction in which the protruding rib protrudes may have an approximately rectangular shape.
(391) That is, the protruding ribs **8312***h***1**, **8312***h***2**, **8312***h***3** may be formed into a rectangular pillar having the same circumferential width and the same radial width and protrude from the lower end **8312***b* of the duct coupling part **8312**.
(392) However, the heights at which the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** protrude downward from the lower end **8312***b* of the duct coupling part **8312** may be set differently.
(393) That is, the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** protrude toward the male screw threads **8514** that extend spirally in different positions. A maximum height of the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** may differ such that a gap between the first, second and third protruding ribs **8312***h***1**, **8312***h***2**, **8312***h***3** and the male screw thread **8514** remains constant in the position of each of the protruding ribs.
(394) For example, the first protruding rib **8312***h***1** may protrude at a first height that is the smallest value, the second protruding rib **8312***h***2** may protrude at a second height greater than the first height, and the third protruding rib **8312***h***3** may protrude at a third height greater than the second height.
(395) The first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3**, as described hereafter, may be disposed clockwise consecutively, when viewed from the upper portion side of the airflow guide **83**.
(396) Additionally, the lower end surfaces of the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** may be provided in the form of an inclined surface, in response to the shape of one side surface of the male screw thread **8514** that extends spirally.
(397) Detailed configurations of the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** are described below with reference to FIGS. **28** and **29**.
(398) Further, at least one upper guide holding projection **8312***g* may be integrally formed on the outer circumferential surface of the duct coupling part **8312** and couple the upper guide **832** described hereafter and the lower guide **831** mutually.
(399) At least one upper guide holding projection **8312***g* may have a lamp surface having a predetermined inclination angle with respect to the outer circumferential surface of the duct coupling part **8312**, and a stepped surface formed approximately perpendicularly with respect to the

outer circumferential surface of the duct coupling part **8312**, such that a coupling based on the downward movement of the upper guide **832** is readily ensured but a separation based on the upward movement of the upper guide **832** is not readily ensured.
(400) Further, the channel guide surface **8313** guiding dry airflow F, having passed through the upper end **8312***a* of the duct coupling part **8312**, to the discharge opening **833** may be formed at the front side of the duct coupling part **8312**.
(401) As illustrated in FIG. **19**, a curved surface of an inclined surface may be formed continuously between the upper end and the lower end of the channel guide surface **8313** to minimize flow loss of dry airflow F and the amount of generated flow noise. The upper end of the channel guide surface **8313** may extend to the approximately same height as the upper end **8312***a* of the duct coupling part **8312**, and the lower end may extend in a curved surface shape or an inclined surface shape to the reference surface **8311***a* of the lower guide **831**.
(402) Specifically, the channel guide surface **8313** may be a concave surface that is provided in a way that the channel guide surface **8313** is surrounded by the rear surface of the duct coupling part **8312** and a boundary wall **8313***d*.
(403) That is, the channel guide surface **8313** may be formed into a concave surface to prevent droplets of wash water, which collide with the first edge wall **8311***b*, then are scattered and come in, from moving to the duct coupling part **8312** and the connection duct part **85**, after the droplets collide with the channel guide surface **8313** and are scattered again.
(404) That is, the shape of the channel guide surface **8313** may be formed such that a gap between the channel guide surface **8313** and the upper end surface **8321** of the upper guide **832** remains big to prevent the rescattered droplets of the wash water from bouncing.
(405) For example, the channel guide surface **8313** may comprise a first inclined surface **8313***a* extending from the reference surface **8311***a* in a direction across the duct coupling part **8312**, a second inclined surface **8313***c* having an extension width much less than the first inclined surface **8313***a*, and a curved surface part **8313***b* disposed between the first inclined surface **8313***a* and the second inclined surface **8313***c*.
(406) The first inclined surface corresponds to a portion occupying most of the surface area of the channel guide surface **8313**. Additionally, the reference surface **8311***a* and the first inclined surface **8313***a* are disposed in succession. To suppress the rescattering of wash water to a maximum degree, the first inclined surface **8313***a* is formed into a flat surface having a minimum climb angle. For example, the climb angle of the first inclined surface **8313***a* may be about 10 degrees or so with respect to the reference surface.
(407) The curved surface part **8313***b* and the second inclined surface **8313***c* are followed by the first inclined surface **8313***a* to prevent a rapid change in the inner flow path in which dry airflow F flows.
(408) The second inclined surface **8313***c*, as illustrated, may be formed into a flat surface, like the first inclined surface **8313***a*, but has a climb angle much greater than that of the first inclined surface **8313***a*.
(409) For example, the climb angle of the second inclined surface **8313***c* may be about 80 degrees or so with respect to the reference surface **8311***a*.
(410) Additionally, the lower guide **831** of the airflow guide **83** of one embodiment may comprise a division wall **8314** that protrudes upward from the upper end **8312***a* of the duct coupling part **8312** as a third means of minimizing the flow of wash water into the airflow guide **83** and the connection duct part **85** through the discharge opening **833**.
(411) The division wall **8314** finally blocks droplets of wash water, which have passed through the above-described blocking rib **8328** and blocking wall **8329** in the state of being scattered after a collision, from entering into the duct coupling part **8312**.
(412) To this end, the division wall **8314**, as illustrated in FIGS. **19**, **20** and **22**, may protrude upward from the upper end **8312***a* of the duct coupling part **8312**, divide the area where the duct

coupling part **8312** is formed and the area where the channel guide surface **8313** is formed, and extend in the form of a barrier that blocks the upper end of the duct coupling part **8312**.
(413) That is, the division wall **8314** may be boundary wall that is disposed between the duct coupling part area and the channel guide surface area and distinguishes the duct coupling part area from the channel guide surface area.
(414) As illustrated, the thickness of the division wall **8314** may remain constant from the right end portion thereof to the left end portion thereof.
(415) At this time, the right end portion of the division wall **8314** may be a free end portion, and the left end portion may extend to the boundary wall **8313***d* and integrally connect to the boundary wall **8313***d*.
(416) Additionally, to minimize flow resistance of dry airflow F moving toward the channel guide surface **8313** past the division wall **8314**, the up-down position of the upper end of the division wall **8314** may remain constant approximately from the right end portion thereof to the left end portion thereof.
(417) However, the up-down position of the upper end of the division wall **8314** may be lower than the upper end **8511** of the duct main body **851** and higher than the lower end edge of the blocking wall **8329** of the upper guide **832** such that the division wall **8314** blocks scattered droplets effectively while avoiding interference with the flow of dry airflow F and preventing a rapid increase in the flow resistance of dry airflow F.
(418) That is, the up-down position of the upper end of the division wall **8314** may be higher than that of the lower end edge **8329***b***3** of the second blocking wall **8329***b* that is in the highest position of the lower end edge of the blocking wall **8329** of the upper guide **832**.
(419) Hereafter, a means of preventing the inflow of wash water, provided at the airflow guide **83** of the dishwasher **1** of one embodiment, is described with reference to FIGS. **23** and **24**.
(420) As describe above, the discharge opening **833** of the airflow guide **83** spraying dry air to the wash space **21** of the tub **20** is open in the wash space **21**.
(421) Additionally, since the dry air supply part **80** is in a non-operation state in the washing stage or the rinsing stage, it is highly likely that wash water is scattered in a droplet state and flows into the inner flow space of the airflow guide **83** through the discharge opening **833**. The droplets of the incoming wash water may also be recondensed in the airflow guide **83**, pass through the duct coupling part **8312** and then flow into the dry air supply part **80**.
(422) To prevent the inflow of wash water, the airflow guide **83** of one embodiment is provided with a means of preventing the inflow of wash water as follows.
(423) The first rib **8328***a* may be provided at the upper end edge, the front edge and the rear edge of the first notch hole **8324** and extend in a shade shape, and at least one second rib **8328***b* may be provided in the first notch hole **8324** and extend across the first notch hole **8324** along the front-rear direction (F-R direction).
(424) By doing so, the inflow of droplets of wash water directly passing through the discharge opening **833** or the first notch hole **8324** may be blocked primarily.
(425) Further, the first blocking wall **8329***a* and the second blocking wall **8329***b* may be provided in the first notch hole **8324**, and extend downward from the lower portion of the upper end surface **8321** of the upper guide **832** to at least partially cover the upper portion side of the discharge opening **833** or the first notch hole **8324**.
(426) Among droplets of wash water that flows into the airflow guide **83** in the state of being scattered through the first notch hole **8324**, droplets bouncing upward collide with the first blocking wall **8329***a* and the second blocking wall **8329***b* and move downward along the first blocking wall **8329***a* and the second blocking wall **8329***b*.
(427) The first blocking wall **8329***a* and the second blocking wall **8329***b*, as illustrated, are disposed in the channel guide surface area. Accordingly, droplets of wash water, which are blocked by the first blocking wall **8329***a* and the second blocking wall **8329***b*, may flow down to the

channel guide surface **8313** by using gravity.
(428) Further, droplets of wash water, which are not blocked by the first blocking wall **8329***a* and the second blocking wall **8329***b*, may be finally blocked from moving toward the duct coupling part **8312** by the division wall **8314** that protrudes upward from the upper end **8312***a* of the duct coupling part **8312**.
(429) As describe above, the up-down position of the upper end of the division wall **8314** is higher than the up-down positions of the lower end edge **8329***a***3** of the first blocking wall **8329***a* and the lower end edge **8329***b***3** of the second blocking wall **8329***b*.
(430) Thus, droplets of wash water having flown into the airflow guide **83** may be blocked from moving to the duct coupling part **8312** while colliding with the first blocking wall **8329***a* and the second blocking wall **8329***b*, but droplets of wash water, which avoid the first blocking wall **8329***a* and the second blocking wall **8329***b* and are scattered toward the duct coupling part **8312** while colliding with the channel guide surface **8313**, may collide with the division wall **8314** without bouncing higher than the upper end of the division wall **8314**.
(431) As described about the first blocking wall **8329***a* and the second blocking wall **8329***b*, droplets of wash water, which collide with the division wall **8314** and are blocked by the division wall **8314**, may flow down to the channel guide surface **8313** along the division wall **8314** by using gravity without moving toward the duct coupling part past the upper end of the division wall **8314**.
(432) As described above, droplets of wash water, blocked by the first blocking wall **8329***a* and the second blocking wall **8329***b*, and droplets of wash water, blocked by the division wall **8314**, may move to the channel guide surface **8313** and naturally be discharged to the bottom tub **20***c* through the discharge opening **833**.
(433) Additionally, the upper end **8511** of the duct main body **851** protrudes to a position higher than the division wall **8314** of the lower guide **831** while protruding upward from the upper end **8312***a* of the duct coupling part **8312** of the lower guide **831**.
(434) Accordingly, even if droplets of washer are produced past the upper end of the division wall **8314**, it is highly likely that the droplets do not reach the upper end **8511** of the duct main body **851** that is disposed in a higher position than the upper end of the division wall **8314** with respect to the up-down direction.
(435) Further, wash water may be collected on the lower surface **25** of the tub **20** at a predetermined water level or above, in the washing stage or the rinsing stage.
(436) An increasing water level of wash water may lead to the flow of wash water into the airflow guide **83** through the discharge opening **833** of the airflow guide **83** and the infiltration of water into a gap between the inner circumferential surface **8312***c* of the duct coupling part **8312** and the outer circumferential surface of the duct main body **851**.
(437) That is, the airflow guide **83** itself is likely to be submerged by wash water.
(438) However, even if the airflow guide **83** is submerged as described above, wash water having flown into the airflow guide **83** may be discharged out of the airflow guide **83** along the channel guide surface **8313** through the discharge opening **833** again in the state where the water level of the wash water does not exceed the upper end of the division wall **8314**.
(439) Additionally, even if the water level of wash water exceeds the upper end of the division wall **8314**, the height of the division wall **8314** remains lower than the height of the upper end **8511** of the duct main body **851**, as described above. Thus, the wash water may not reach the upper end **8511** of the duct main body **851**, and the wash water having arrived at the upper end **8312***a* of the duct coupling part **8312** may be discharged from the upper end **8312***a* of the duct coupling part **8312** to the lower surface **25** of the tub **20** again, through the gap between the inner circumferential surface **8312***c* of the duct coupling part **8312** and the outer circumferential surface of the duct main body **851**.
(440) By setting the height of the division wall **8314**, the height of the upper end **8312***a* of the duct coupling part **8312** and the height of the upper end **8511** of the duct main body **851**, wash water

may be prevented from flowing into the heater housing **81** and the heater effectively past the upper end **8511** of the duct main body **851** even if the wash water is scattered ad flows into the airflow guide **83** or submerges the airflow guide **83**.
(441) Further, the airflow guide **83** of the dishwasher **1** of one embodiment may help to minimize resistance against the flow of dry airflow F supplied through the upper end **8511** of the duct main body **851** of the connection duct part **85** while blocking and minimizing the movement of droplets of wash water coming in through the first notch hole **8324** and the discharge opening **833** to the duct coupling part **8312** and the duct main body **851**, based on the positions of the blocking rib **8328** of the upper guide **832**, the blocking wall **8329**, the division wall **8314** of the lower guide **831**, and the upper end **8511** of the duct main body **851**.
(442) Specifically, since the height of the upper end of the division wall **8314** remains lower than the upper end **8511** of the duct main body **851** into which dry airflow F flows, as illustrated in FIG. **24**, the division wall **8314**'s resistance against the flow of dry airflow F having passed through the upper end **8511** of the duct main body **851** may be minimized.
(443) Further, the flow path of dry airflow F having passed through the upper side of the division wall **8314** is partially blocked by the first blocking wall **8329***a* and the second blocking wall **8329***b*, but the lower end edge **8329***a***3** of the first blocking wall **8329***a* and the lower end edge **8329***b***3** of the second blocking wall **8329***b* are spaced upward from the channel guide surface **8313** by a predetermined distance.
(444) Accordingly, the dry airflow F may flow effectively through a space between the lower end edge **8329***a***3** of the first blocking wall **8329***a* and the channel guide surface **8313** and a space between the lower end edge **8329***b***3** of the second blocking wall **8329***b* and the channel guide surface **8313**.
(445) Further, the other end edge **8329***a***2** of the first blocking wall **8329***a* and the other end edge **8329***b***2** of the second blocking wall **8329***b* may be respectively separated from the outer wall surface of the upper guide **832**.
(446) Thus, dry airflow F may flow effectively through a space between the other end edge **8329***a***2** of the first blocking wall **8329***a* and the outer wall surface of the upper guide **832** and a space between the other end edge **8329***b***2** of the second blocking wall **8329***b* and the outer wall surface of the upper guide **832**.
(447) The above-described configurations of the first blocking wall **8329***a* and the second blocking wall **8329***b* may help to minimize an increase in the flow resistance against dry airflow F and maximize a flow path of dry airflow F.
(448) [Assembly of Airflow Guide and Means of Limiting Relative Movement]
(449) Hereafter, the process of assembling and fixing the airflow guide **83** of one embodiment to the connection duct part **85** and a means of limiting movement relative to the connection duct part are described with reference to FIGS. **25**A to **27**B.
(450) As described above, the lower guide **831** may be coupled to the duct main body **851** of the connection duct part **85** based on a two-stage coupling manipulation. Preferably, the two-stage coupling manipulation may comprise an up-down simple perpendicular movement manipulation and a circumferential simple rotational movement manipulation.
(451) For the up-down perpendicular movement manipulation, in the state where the cap cover **834** and the upper guide **832** are coupled to the lower guide **831**, the airflow guide **83** may be arranged to be disposed at the upper side of the duct main body **851** of the connection duct part **85**, as illustrated in FIGS. **25**A and **25**B.
(452) At this time, the central axes of the duct main body **851** and the duct coupling part **8312** may be aligned in the up-down direction (U-D direction) to be inserted into the lower end **8312***b* of the duct coupling part **8312** of the lower guide **831** of the upper end **8511** of the duct main body **851**.
(453) Additionally, the airflow guide **83** may be rotated clockwise around the duct coupling part **8312** from the fixed position within a predetermined range of angles. The position of the airflow

guide **83** rotated clockwise is a position in which the guide projection **8516** formed on the outer circumferential surface of the duct main body **851** can be inserted into the first guide groove **8312***d* of the duct coupling part **8312**.

(454) When the arrangement of the airflow guide **83** is completed with respect to the duct main body **851** as illustrated, the airflow guide **83** moves perpendicularly in the downward direction (D-direction) along the direction indicated by the arrow in FIGS. **25**A and **25**B such that the upper end **8511** of the duct main body **851** is inserted into the lower end **8312***b* of the duct coupling part **8312**. Accordingly, the guide projection **8516** of the duct main body **851** may be inserted into the lower end portion of the first guide groove **8312***d*.

(455) In the state where the guide projection **8516** of the duct main body **851** is inserted into the lower end portion of the first guide groove **8312***d* as described above, as the airflow guide **83** perpendicularly moves in the downward direction (D-direction), the up-down perpendicular movement manipulation of the airflow guide **83** starts.

(456) Accordingly, the movement of the first guide groove **8312***d* of the duct coupling part **8312** is guided by the guide projection **8516** that stands still, and the first guide groove **8312***d* perpendicularly moves in the downward direction (D-direction).

(457) As the guide projection reaches the upper end of the first guide groove **8312***d* as illustrated in FIGS. **26**A and **26**B, the airflow guide **83** may not perpendicularly move in the downward direction (D-direction) any longer, based on the guide projection **8516**'s action.

(458) At this time, since the guide projection **8516** reaches one end portion of the second guide groove **8312***e*, the airflow guide **83** may not make a perpendicular movement in the downward direction (D-direction), but may make a rotational movement circumferentially along the direction indicated by the arrow in FIG. **26**A.

(459) For the circumferential rotational movement manipulation of the two-stage coupling manipulation, as the airflow guide **83** rotates counterclockwise along the direction indicated by the arrow, the second guide groove **8312***e* may start to make a rotational movement counterclockwise along the guide projection **8516** that sands still.

(460) As the airflow guide **83** starts to make a rotational movement counterclockwise, the guide projection **8516** reaches a stopper projection **8312***f* that is disposed near the other end portion of the second guide groove **8312***e*.

(461) At this time, stick-slip in relation to the rotation of the airflow guide **83** may be formed by the stopper projection **8312***f*, and as a rotational force is additionally applied, the guide projection **8516** may go over the stopper projection **8312***f*.

(462) Then as the guide projection **8516** reaches the other end portion of the second guide groove **8312***e* past the stopper projection **8312***f*, the airflow guide **83** may not rotate counterclockwise any longer because of the guide projection **8516**'s action.

(463) When the airflow guide **83** cannot rotate any longer as described above, the coupling between the lower guide **831** and the duct main body **851** may be completed, and as long as another external force is not applied, the guide projection may be fixed to the other end portion side of the second guide groove **8312***e* by the stopper projection **8312***f*, as illustrated in FIGS. **27**A and **27**B.

(464) Accordingly, the airflow guide **83** may be disposed in a fixed position of the connection duct part **85**, based on a very simple manipulation or assembly process comprising the simple perpendicular movement manipulation and the simple rotational movement manipulation.

(465) However, as the arrangement of the airflow guide **83** is completed, the self weight-induced movement of the airflow guide **83** is impossible, but as strong external force is applied, the second guide groove **8312***e* and the guide projection **8516** are likely to be easily unheld from each other.

(466) Further, the lower guide **831** and the airflow guide **83** are highly likely to clatter because of a gap between the duct coupling part **8312** of the lower guide **831** and the duct main body **851** of the connection duct part **85**, caused by manufacturing tolerance.

(467) To prevent the second guide groove **8312***e* and the guide projection **8516** from being unheld

from each other and keep them fixed in their fixed positions, the release prevention part **8311***e* may be provided.
(468) To prevent a relative rotation in an elastic hook way, the release prevention part **8311***e*, as described above, may be provided with a hook part that is integrally formed on the second edge wall **8311***c* of the lower guide **831** in a way that a L-shaped notch is formed at the lower end side of the second edge wall **8311***c* of the lower guide **831**, i.e., in a way that the lower end of the second edge wall **8311***c* is partially cut.
(469) Preferably, the hook part constituting the release prevention part **8311***e* may be disposed near the left edge side of the guide main body **8311** of the lower guide **831**, on the second edge wall **8311***c*.
(470) Specifically, the release prevention part **8311***e* may be formed into a circular arc-shaped plate that is disposed in a way that surrounds the circumference of the fastening nut **852** disposed inside the second edge wall **8311***c*, in a circular arc shape, and is elastically deformable, as illustrated in FIGS. **28** and **29**.
(471) One end portion of the release prevention part **8311***e* functions as a fixation end portion **8311***e***1** integrally connecting to the second edge wall **8311***c*. As described hereafter, the fixation end portion **8311***e***1** may also provide resilient force or elastic force to the other end portion side that is deformed at a time of rotational movement manipulation for installing and fixing the airflow guide **83**.
(472) The other end portion of the release prevention part **8311***e* separates from the second edge wall **8311***c* and functions as a free end portion **8311***e***2**, and directly contacts one lateral surface of the stopper **8521** provided in the upper portion of the fastening nut **852** to stop the rotation of the lower guide **831**.
(473) To keep the free end portion **8311***e***2** contacting and holding the stopper **8521** directly and effectively, the radial thickness of the free end portion **8311***e***2** may be greater than the radial thickness of the fixation end portion **8311***e***1**.
(474) Specifically, as illustrated, a distance between an outer circumferential surface **8311***e***4** of the release prevention part **8311***e* and the central axis C of the duct coupling part **8312** remains constant, and an inner circumference surface **8311***e***3** of the release prevention part **8311***e* may comprise a portion where a distance from the central axis C of the duct coupling part **8312** remains constant, and a portion where a distance from the central axis C of the duct coupling part **8312** changes.
(475) As illustrated, the inner circumferential surface **8311***e***3** of the release prevention part **8311***e* may comprise a non-contact surface that stays in no contact with the radial outer end portion of the stopper **8521**, and a contact surface that is in contact with the radial outer end portion of the stopper **8521**, for example.
(476) The non-contact surface extends toward the free end portion **8311***e***2** from the fixation end portion **8311***e***1**, has the same curvature as the outer circumferential surface **8311***e***4** and corresponds to a portion where a distance from the central axis C of the duct coupling part **8312** remains constant.
(477) The contact surface extends to the free end portion **8311***e***2** from the position where the non-contact surface ends and corresponds to a portion where a distance from the central axis C of the duct coupling part **8312** decreases gradually.
(478) The free end portion **8311***e***2** corresponding to the position where the contact surface ends may protrude further inward than the radial outer end portion of the stopper **8521** in the state where the release prevention part **8311***e* is not deformed as illustrated in FIG. **28**.
(479) Additionally, the contact surface corresponds to a portion that is directly pressurized by the radial outer end portion of the stopper **8521** while the lower guide **832** rotates.
(480) Accordingly, the contact surface may be formed into a curved surface or an inclined surface such that the relative movement of the radial outer end portion of the stopper **8521** is smoothly

made and frictional force decreases, for example.
(481) Further, a tool groove **8311***e***5** may be formed on the end portion surface of the free end portion **8311***e***2** of the release prevention part **8311***e* and be concave toward the fixation end portion **8311***e***1** side with respect to the circumferential direction.
(482) In the state where the airflow guide **83** is fixed in the fixed position completely, even if external force is applied, the airflow guide **83** is configured not to rotate in a direction where the airflow guide **83** separates from the connection duct part **85** as long as the release prevention part **8311***e* or the stopper **8521** is not broken.
(483) Additionally, the airflow guide **83** is formed in a position farthest from the front surface of the tub **20**, and the release prevention part **8311***e* is formed in a position facing the left side surface of the bottom tub **20***c*. The positions are hardly reached by the user, and the user cannot undo the holding state between the release prevention part **8311***e* and the stopper **8521** easily without an additional tool.
(484) For the user to easily deform the release prevention part **8311***e* and undo the holding state between the free end portion **8311***e***2** of the release prevention part **8311***e* and the stopper **8521** with an ordinary tool such as a screwdriver and the like, the tool groove **8311***e***5** may be provided on the end portion surface of the free end portion **8311***e***2** of the release prevention part **8311***e*.
(485) At this time, the tool groove **8311***e***5** may have a polygonal cross section as illustrated for an ordinary tool to be readily held at a time of undoing the holding state. FIGS. **28** and **29** show an embodiment comprising a tool groove **8311***e***5** having a cross section of a ⊏ shape, among polygons, for example.
(486) Additionally, the release prevention part **8311***e* is configured to repeat elastically deformation at a time of assembling and separating the airflow guide **83**.
(487) Repetitive deformation may result in fatigue fracture. To prevent fatigue fracture, at least one reinforcement rib **8311***e***6** may be integrally provided on the outer circumferential surface **8311***e***4** of the release prevention part **8311***e* and protrude outward in the radial direction.
(488) Hereafter, a relationship between the stopper **8521** of the fastening nut **852** and the operation of the release prevention part **8311***e* in the coupling process of the airflow guide **83** is described.
(489) As described above, the lower guide **831** of the airflow guide **83** may be coupled to the duct main body **851** of the connection duct part **85** based on the two-stage coupling manipulation comprising a simple up-down perpendicular movement manipulation and a simple circumferential rotational movement manipulation.
(490) As the up-down perpendicular movement manipulation is completed, the free end portion **8311***e***2** of the release prevention part **8311***e* may be disposed in a space between a pair of stoppers **8521** that are adjacent to each other with respect to the circumferential direction, as illustrated in FIG. **28**.
(491) As illustrated, the space between the pair of adjacent stoppers **8521** may be embodied by a circular arc groove that is open toward the outside in the upward direction and the radial direction.
(492) As the circumferential rotational movement manipulation starts after the up-down perpendicular movement manipulation is completed, out of the pair of stoppers **8521**, the radial outer end portion of a stopper **8521** disposed forward with respect to the rotation direction of the airflow guide **83**, and the contact surface of the release prevention part **8311***e* may start to contact each other.
(493) As the radial outer end portion of the stopper **8521** and the contact surface of the release prevention part **8311***e* start to contact, the rotational force of the airflow guide **83** is converted into a pressurizing force against the release prevention part **8311***e*.
(494) Accordingly, the release prevention part **8311***e* is pressurized by the stopper **8521** that stands still, and is pushed gradually outward in the radial direction from an initial position that is no load state and starts to be elastically deformed.
(495) At this time, the release prevention part **8311***e* may be elastically deformed continuously to

the position where the free end portion **8311***e***2**'s contact with the radial outer end portion of the stopper **8521** is undone.
(496) Thus, as illustrated in FIG. **29**, as the contact between the free end portion **8311***e***2** of the release prevention part **8311***e* and the stopper **8521** is undone, the release prevention part **8311***e* may instantly return to the initial position with a click by using elasticity.
(497) The free end portion **8311***e***2** of the release prevention part **8311***e* protrudes to the inside of a circular arc groove formed at the front side of the stopper **8521** at the same time as the release prevention part **8311***e* returns to the initial position.
(498) Accordingly, once the free end portion **8311***e***2** of the release prevention part **8311***e* enters into the circular arc groove, based on the counterclockwise rotation of the airflow guide **83**, as illustrated, the clockwise rotation opposite to the counterclockwise rotation must be limited by the lateral surface of the stopper **8521**.
(499) By doing so, a rotational movement in a direction where the airflow guide **83** separates from the duct main body **851** of the connection duct part may be prevented effectively through the release prevention part **8311***e* and the stopper **8521**.
(500) However, in the state where a rotational movement is not completed during the two-stage coupling manipulation, the free end portion **8311***e***2** of the release prevention part **8311***e* may rotate toward another adjacent stopper **8521** past the previous stopper **8521**. At this time, the free end portion **8311***e***2** of the release prevention part **83711***e* may continue to rotate further while going over another stopper **8521**, in the same way described above, and the airflow guide **83**'s rotation in the opposite direction may be limited in the same way.
(501) The additional rotational movement may be performed until the guide projection provided at the duct main body **851** reaches the other end portion of the second guide groove **8312***e*, as described above.
(502) Additionally, a relative movement, in particular, a downward relative movement, of the airflow guide **83**, caused by a gap between the second guide groove **8312***e* and the guide projection **8516**, may be limited and minimized through the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3** that protrude downward from the lower end **8312***b* of the duct coupling part **8312**.
(503) FIG. **30** shows that the airflow guide **83** is arranged in a fixed position completely. For convenience, the cap cover **834** and the upper guide **832** are omitted in FIG. **30**.
(504) As the fixation and arrangement of the airflow guide **83** are completed as illustrated in FIG. **30**, a gap having a predetermined width may be formed between the first to third protruding ribs **8312***h***1**, **8312***h***2**, **8312***h***3** and the male screw thread **8514** of duct main body **851** respectively.
(505) As illustrated in the cross-sectional views of FIGS. **32** to **34**, the first protruding rib **8312***h***1**, the second protruding rib **8312***h***2** and the third protruding rib **8312***h***3**, disposed at regular intervals around the circular opening formed at the lower end **8312***b* of the duct coupling part **8312**, may have a different maximum protrusion height such that the gap between the protruding ribs and the male screw thread **8514** is maintained at a predetermined level or below in the position of each of the protruding ribs.
(506) At this time, a gap between the first protruding rib **8312***h***1** and the male screw thread **8514** of the duct main body **851**, a gap between the second protruding rib **8312***h***2** and the male screw thread **8514** of the duct main body **851**, and a gap between the third protruding rib **8312***h***3** and the male screw thread **8514** of the duct main body **851** may differ depending on manufacturing tolerance of each of the protruding ribs.
(507) However, the entire amount of a downward relative movement of the airflow guide **83** may be limited based on a gap having a maximum width among the gaps.
(508) At this time, the maximum width of a gap may be 0.01 mm or less, for example.
(509) Since the maximum gap is limited to 0.01 mm or less, the user may not recognize a clatter caused by the amount of relative movement or displacement corresponding to 0.01 mm or less even

though each of the protruding ribs **8312**_h_**1**, **8312**_h_**2**, **8312**_h_**3** does not directly contact the male screw thread **8514** of the duct main body **851**.
(510) Further, as illustrated in FIG. **31**, the male screw thread **8514** of the duct main body **851**, when viewed from above, may be provided as a right-handed screw to be tightened clockwise.
(511) Thus, the male screw thread **8514** extends in a spiral shape the up-down position of which gradually becomes low clockwise, when viewed from above.
(512) When viewed from above, the first protruding rib **8312**_h_**1**, the second protruding rib **8312**_h_**2** and the third protruding rib **8312**_h_**3** may be consecutively disposed clockwise such that the gap between each of the protruding ribs **8312**_h_**1**, **8312**_h_**2**, **8312**_h_**3** and the male screw thread **8541** of the duct main body **851** remains similar approximately.
(513) That is, the protrusion heights of the protruding ribs **8312**_h_**1**, **8312**_h_**2**, **8312**_h_**3** may gradually become high clockwise in response to the position of the male screw thread **8514** of the duct main body **851** that gradually becomes low clockwise.
(514) Accordingly, the counterclockwise rotational movement manipulation starts in the state where the perpendicular movement manipulation is completed, the gap between each of the protruding ribs **8312**_h_**1**, **8312**_h_**2**, **8312**_h_**3** and the male screw thread **8514** of the duct main body **851** may decrease gradually.
(515) Additionally, as illustrated in FIG. **31**, the male screw thread **8514** of the duct main body **851** may have at least three or more windings from an upper end **8514**_b_ thereof to a lower end **8514**_a_ thereof, for example.
(516) At this time, as illustrated in FIGS. **32** and **33**, a screw thread **8523**_a_ of the fastening nut **852** to be screw-coupled to the male screw thread **8514** of the duct main body **851** may have two windings that are less than those of the male screw thread **8514** of the duct main body **851**, for example.
(517) Thus, as a screw coupling of the fastening nut **852** to the male screw thread **8514** of the duct main body **851** is completed, the upper side surface of the screw thread of the male screw thread **8514**, corresponding to one winding from the upper end **8514**_b_ of the male screw thread **8514** of the duct main body **851**, may remain exposed to the lower end **8312**_b_ of the duct coupling part **8312**.
(518) As a result, the first protruding rib **8312**_h_**1**, the second protruding rib **8312**_h_**2** and the third protruding rib **8312**_h_**3** of the duct coupling part **8312** may be supported by the upper side surface of the male screw thread **8514** of the duct main body **851** at a time of occurrence of a downward relative movement, without being affected by the screw thread **8523**_a_ provided on an inner circumferential surface **8523** of the fastening nut **852**.
(519) Further, the radial maximum widths of the first protruding rib **8312**_h_**1**, the second protruding rib **8312**_h_**2** and the third protruding rib **8312**_h_**3** may be the same, and be less than a gap between the outer circumferential surface of the connection duct part **85** and the inner circumferential surface **8523** of the fastening nut **852**.
(520) Accordingly, the first protruding rib **8312**_h_**1**, the second protruding rib **8312**_h_**2** and the third protruding rib **8312**_h_**3** may be in no contact with the inner circumferential surface **8523** and the upper end **8522** of the fastening nut **852**.
(521) By doing so, although the third protruding rib **8312**_h_**3** having a maximum protrusion height is inserted into the fastening nut **852**, a relative position of the third protruding rib **8312**_h_**3** with respect to the male screw thread **8514** of the duct main body **851**, as illustrated in FIG. **33**, may be set regardless of the inner circumferential surface **8523** and the upper end **8522** of the fastening nut **852**.
(522) At this time, as described above, the lower end surfaces of the first protruding rib **8312**_h_**1**, the second protruding rib **8312**_h_**2** and the third protruding rib **8312**_h_**3** may be provided in the form of an inclined surface to correspond to the shape of the upper side surface of the male screw thread **8514** of the duct main body **851**, which extends spirally.

(523) The inclination angles of the lower end surfaces of the first protruding rib **8312*h*1**, the second protruding rib **8312*h*2** and the third protruding rib **8312*h*3** may be approximately the same as that of the male screw thread **8514** of the duct main body **851**.
(524) Thus, as at least any one of the first protruding rib **8312*h*1**, the second protruding rib **8312*h*2** and the third protruding rib **8312*h*3** contacts the upper side surface of the male screw thread **8514** of the duct main body **851** because of the airflow guide **83**'s downward relative movement caused by external force, at least any one of the first protruding rib **8312*h*1**, the second protruding rib **8312*h*2** and the third protruding rib **8312*h*3** may be in surface-contact with the upper side surface of the male screw thread **8514** of the duct main body **851**.
(525) That is, the lower end surface of at least any one of the first protruding rib **8312*h*1**, the second protruding rib **8312*h*2** and the third protruding rib **8312*h*3** may be entirely supported by the upper side surface of the male screw thread **8514** of the duct main body **851**.
(526) Since a maximum relative contact surface between the lower end surface of at least any one of the protruding ribs and the male screw thread **8514** of the duct main body **851** is ensured, the downward relative movement of the airflow guide **83** may be effectively limited, and the airflow guide **83** may be reliably supported, despite strong external force.
(527) [Detailed Configuration of Fastening Nut]
(528) Hereafter, a detailed configuration of the fastening nut **852** provided in the dishwasher **1** of one embodiment is described with reference to FIGS. **35** and **36**.
(529) The fastening nut, as described above, is screw-coupled to the outer circumferential surface of the connection duct part **85** and fixes the connection duct part **85** to the lower surface **25** of the bottom tub **20*c***.
(530) To this end, the fastening nut **852** may have a cylindrical shape entirely, and be provided with a screw thread **8523*a*** on the cylinder-shaped inner circumferential surface **8523*a*** thereof, and the screw thread **8523*a*** has a female screw shape and is screw-coupled to the male screw thread **8514** of the duct main body **851** of the connection duct part **85**.
(531) The screw thread **8523*a*** of the fastening nut, as described above, may have two windings, for example.
(532) At a time of fixing and coupling the duct main body **851**, the fastening nut **852** is screw-coupled to the male screw thread **8514** of the duct main body **851** such that the upper end **8511** of the duct main body **851** is fixed to the tub **20**, in the state of being exposed to the inside of the tub **20**.
(533) At this time, in the state where the fastening nut **852** is in close contact with the upper portion side of the lower surface **25** of the bottom tub **20*c***, and a flange **8513** of the duct main body **851** is in close contact with the lower portion side of the lower surface **25** of the bottom tub **20*c***, the flange **8513** receives the force of being pulled toward the lower surface of the bottom tub **20*c***, because of the fastening nut **852**'s coupling force.
(534) By doing so, adhesive force between the flange **8513** and the lower surface **25** of the bottom tub **20*c*** increases. Thus, it is less likely that wash water leaks to the outer circumferential surface of the duct main body **851**. As a means of promoting the effect of preventing the leakage of wash water, an airtight ring made of an elastic material may be further provided between the flange **8513** and the lower surface **25** of the bottom tub **20*c***.
(535) Additionally, a plurality of stoppers **8521** may be provided at the upper end side of the fastening nut **852** and interact with the release prevention part **8311*e*** of the airflow guide **83**.
(536) The plurality of stoppers **8521** may be formed in a way that the upper end **8522** and the outer circumferential surface **8524** of the fastening nut **852** are depressed partially, and each of the plurality of stoppers **8521** may be disposed along the circumferential direction at regular intervals.
(537) As described above, the other end portion of the release prevention part **8311*e*** of the airflow guide **83** separates from the second edge wall **8311*c*** and functions as a free end portion **8311*e*2**, and directly contacts one side surface of the stopper **8521** of the fastening nut **852** and limits the

rotational movement of the airflow guide **83** to prevent the airflow guide **83**'s escape from the right position thereof.
(538) Further, the fastening nut **852** is directly exposed to wash water in the washing stage and the rinsing stage. In particular, the fastening nut **852** is directly exposed to the wash pace, and in some cases, is submerged in wash water filling the lower surface **25** of the bottom tub **20***c*, since it is disposed at the upper side of the lower surface **25** of the bottom tub **20***c*.
(539) To be protected from corrosion caused by wash water, the fastening nut **852** may be manufactured, based on a plastic injection molding process.
(540) Further, as the fastening nut **852** is coupled to the duct main body **851** completely, as described above, a lower end surface **8525** of the fastening nut **852** is in close contact with the lower surface **25** of the bottom tub **20***c* directly.
(541) At this time, a minute gap may be formed between the lower surface **25** of the bottom tub **20***c* and the lower end surface **8525** of the fastening nut **852**, in close contact, due to their manufacturing tolerance, and through the gap, wash water may flow into a space formed by the inner circumferential surface **8523** of the fastening nut **852**, the outer circumferential surface of the duct main body **851** and the lower surface **25** of the bottom tub **20***c*.
(542) Since the gap is minute, the wash water can hardly be discharged out of the fastening nut **852** once the wash water flows into the space.
(543) The lower surface **25** of the bottom tub **20***c* is highly likely to be corroded by the wash water which is drawn but not discharged. In particular, since the dry air supply hole **254** is formed inside the fastening nut **852** in a way that the bottom tub **20***c* is perforated, the dry air supply hole **254** is highly likely to corrode, and germs included in wash water are reproduced generating a bad smell.
(544) As a means of preventing the bottom tub **20***c′* corrosion and generation of a bad smell, the fastening nut **852** provided at the dishwasher **1** of one embodiment may comprise a plurality of contact projections **8526** that extends toward the bottom tub **20***c* from the lower end surface **8525**.
(545) Each of the plurality of contact projections **8526** may have an upper end that integrally connects to the lower end surface **8525** of the fastening nut **852**, and a lower end that extends toward the lower surface **25** of the bottom tub **20***c* in the form of a protruding rib.
(546) Each of the plurality of contact projections **8526** may be spaced from one another by a predetermined circumferential distance Dc, on the lower end surface **8525** of the fastening nut **852**, in the circumferential direction.
(547) At this time, the predetermined circumferential distance may be greater than a maximum circumferential thickness Wc of each contact projection **8526**.
(548) By doing so, a wash water passage through which wash water comes in and out may be provided between the contact projections **8526**, and the lower end surface **8525** of the fastening nut **852** may be separated from the lower surface **25** of the bottom tub **20***c* and exposed to the wash space.
(549) Accordingly, wash water may be effectively discharged out of the fastening nut without staying in the space formed by the inner circumferential surface **8523** of the fastening nut **852**, the outer circumferential surface of the duct main body **851** and the lower surface **25** of the bottom tub **20***c*.
(550) By doing so, the corrosion of the bottom tub **20***c*, i.e., the corrosion of the dry air supply hole **254**, may be prevented effectively.
(551) Further, each contact projection **8526** may have the same outer shape.
(552) At this time, each contact projection **8526** may be formed into a sharp edge having a horizontal cross section of which decreases gradually from an upper end to a lower end, for example.
(553) Additionally, a contact end portion **8526***a* at the lower end of the contact projection **8526**, contacting the lower surface **25** of the bottom tub **20***c*, may be formed into a curved surface that is convex toward the lower surface **25** of the bottom tub **20***c*. Accordingly, the contact projection

8526 and the bottom tub **20***c* may remain at least in a linear contact state.
(554) The shape of the contact projection may help to distribute the fastening nut **852**'s coupling force or pressurizing force uniformly through the each of the contact projections **8526** and apply the same to the lower surface **25** of the bottom tub **20***c*. Further, the sharp edge shape of the contact projection may help to minimize foreign substances such as food and the like fitted or fixed between the contact projection **8526** and the bottom tub **20***c*.
(555) Further, as illustrated in FIG. **36**, the dry air supply hole **254** through which the upper end **8511** of the duct main body **851** passes, is formed on a convergence surface **251** for guiding wash water to the sump hole **252**.
(556) The convergence surface **251** has a predetermined convergence inclination angle with respect to the horizontal direction to allow wash water to be moved by gravity.
(557) Accordingly, the pressurizing force of the fastening nut **852** that moves perpendicularly at a time of screw-coupling the fastening nut **852** to the male screw thread **8514** of the duct main body **851** may not be applied uniformly through the contact projection **8526**.
(558) For the pressurizing force of the contact projection **8526** of the fastening nut **852** to be uniformly distributed and applied to the bottom tub **20***c*, a ring-type coupling surface **2541** may be formed around the dry air supply hole **254** and be pressurized by the contact end portion **8526***a* of the lower end of the contact projection **8526**.
(559) At this time, the ring-type coupling surface **2541** may have directionality that extends in the horizontal direction perpendicular to the perpendicular direction where the fastening nut **852** is moved while being screw-coupled.
(560) For example, the ring-type coupling surface **2541** may be a ring-type bead surface that is formed in a way that the surrounding area of the dry air supply hole **254** is pressed and plastic-deformed.
(561) As described above, since the coupling surface **2541** extending horizontally is additionally formed in the portion of the bottom tub **20***c*, where the pressuring force of the fastening nut **852** is directly applied, each contact projection **8526** may apply its pressurizing force to the bottom tub **20***c* uniformly or evenly.
(562) Further, the bottom tub **20***c*, as illustrated, may be further provided with a cylindrical part **2542** that extends circumferentially along the dry air supply hole **254** and protrudes upward toward the lower end surface **8525** of the fastening nut **852**.
(563) The cylindrical part **2542** intends to extend upward the height of a flooding water surface of wash water filling the lower surface **25** of the bottom tub **20***c*.
(564) As illustrated, to prevent interference with the fastening nut **852**, the height at which the cylindrical part **2542** protrudes from the lower surface **25** of the bottom tub **20***c* may be less than the height at which the contact projection **8526** protrudes from the lower end surface **8525** of the fastening nut **852**.
(565) Additionally, the height at which the cylindrical part **2542** protrudes from the lower surface **25** of the bottom tub **20***c* may remain constant along the circumferential direction.
(566) Thus, it is less likely that wash water flows into the dry air supply hole **254** directly and that the dry air supply hole **254** and the cylindrical part **2542** are corroded by wash water.
(567) The embodiments are described above with reference to a number of illustrative embodiments thereof. However, embodiments are not limited to the embodiments and drawings set forth herein, and numerous other modifications and embodiments can be drawn by one skilled in the art within the technical scope of the disclosure. Further, the effects and predictable effects based on the configurations in the disclosure are to be included within the scope of the disclosure though not explicitly described in the description of the embodiments.

Claims

1. A dishwasher, comprising: a tub that defines a wash space configured to accommodate a wash target, the tub having a front surface that is open; a dry air supply configured to generate dry air for drying the wash target; and an airflow guide configured to guide the dry air generated by the dry air supply to the wash space, wherein the dry air supply comprises: a connection duct that extends upward and passes through a lower surface of the tub, the connection duct being coupled to the airflow guide and configured to supply the dry air into the airflow guide, and a movement limiter configured to, based on the airflow guide being coupled to the connection duct, limit a rotational movement and a downward movement of the airflow guide relative to the connection duct, wherein the airflow guide comprises a duct coupling part that receives an upper end of the connection duct, and wherein the movement limiter comprises: a screw thread provided at the connection duct and configured to limit the downward movement of the duct coupling part, and protruding ribs that are provided at the airflow guide and protrude downward from a lower end of the duct coupling part toward the screw thread.
2. The dishwasher of claim 1, wherein the movement limiter is configured to, based on the duct coupling part being coupled to the connection duct, define a gap between the screw thread and a lower end surface of the at least one of the protruding ribs.
3. The dishwasher of claim 2, wherein the duct coupling part defines an opening that receives the upper end of the connection duct, and wherein the protruding ribs are arranged around the opening and spaced apart from one another.
4. The dishwasher of claim 3, wherein protruding heights of the protruding ribs from the lower end of the duct coupling part are different from one another.
5. The dishwasher of claim 3, wherein circumferential widths of the protruding ribs are equal to one another.
6. The dishwasher of claim 4, wherein the protruding ribs comprise: a first protruding rib that protrudes from the lower end of the duct coupling part defines a first protruding height; a second protruding rib that protrudes from the lower end of the duct coupling part defines a second protruding height greater than the first protruding height; and a third protruding rib that protrudes from the lower end of the duct coupling part defines a third protruding height greater than the second protruding height, and wherein the first protruding rib, the second protruding rib, and the third protruding rib are arranged along the lower end of the connection duct in a clockwise direction or a counterclockwise direction.
7. The dishwasher of claim 6, wherein each of the first protruding rib, the second protruding rib, and the third protruding rib has a lower end surface that protrudes from the lower end of the duct coupling part and is inclined with respect to the lower end of the duct coupling part, and wherein a height of the lower end surface of each of the first protruding rib, the second protruding rib, and the third protruding rib increases along the clockwise direction or the counterclockwise direction.
8. The dishwasher of claim 1, wherein the duct coupling part is configured to be coupled to the connection duct based on a two-stage coupling structure.
9. The dishwasher of claim 8, wherein the two-stage coupling structure is configured to enable (i) a downward movement of the airflow guide toward the connection duct and (ii) a rotational movement of the airflow guide in a first circumferential direction subsequent to the downward movement.
10. The dishwasher of claim 9, wherein the airflow guide is configured to, based on rotating in the first circumferential direction, decrease a gap between the screw thread and a lower end surface of the at least one of the protruding ribs.
11. The dishwasher of claim 10, wherein the dry air supply further comprises a fastening nut disposed at the connection duct, the fastening nut being screw-coupled to the screw thread of the connection duct based on rotating in a second circumferential direction opposite to the first circumferential direction.

12. The dishwasher of claim 9, wherein the movement limiter further comprises: a first guide groove that is defined at an inner circumferential surface of the duct coupling part and extends in an up-down direction; a second guide groove that is connected to an upper end of the first guide groove and extends in a circumferential direction of the duct coupling part; and a guide projection that is disposed at an outer circumferential surface of the connection duct and protrudes toward the inner circumferential surface of the duct coupling part, the guide projection being configured to move along the first guide groove and the second guide groove based on the duct coupling part being coupled to the connection duct.
13. The dishwasher of claim 8, wherein the movement limiter further comprises a release prevention part configured to fix a position of the airflow guide after the rotational movement of the airflow guide.
14. The dishwasher of claim 13, wherein the release prevention part is configured to restrict the airflow guide from rotating in a direction opposite to the rotational movement.
15. The dishwasher of claim 14, wherein the dry air supply further comprises a fastening nut that is screw-coupled to the connection duct and fixes the connection duct to the tub, the fastening nut comprising a plurality of stoppers that are configured to restrict the airflow guide from rotating in the direction opposite to the rotational movement, and wherein the release prevention part comprises a hook that is disposed at the airflow guide and protrudes toward at least one of the plurality of stoppers.
16. A dishwasher, comprising: a tub that defines a wash space configured to accommodate a wash target, the tub having a front surface that is open; a dry air supply configured to generate dry air for drying the wash target; and an airflow guide configured to guide the dry air generated by the dry air supply to the wash space, wherein the dry air supply comprises: a connection duct that extends upward and passes through a lower surface of the tub, the connection duct being coupled to the airflow guide and configured to supply the dry air into the airflow guide, and a movement limiter configured to, based on the airflow guide being coupled to the connection duct, limit a rotational movement and a downward movement of the airflow guide relative to the connection duct, wherein the connection duct comprises a screw thread, wherein the dry air supply further comprises a fastening nut that is screw-coupled to the screw thread of the connection duct and fixes the connection duct to the lower surface of the tub, and wherein a lower end surface of the fastening nut is exposed to the wash space.
17. The dishwasher of claim 16, wherein the fastening nut comprises a plurality of contact projections that extend from the lower end surface of the fastening nut toward the lower surface of the tub.
18. The dishwasher of claim 17, wherein the plurality of contact projections comprise: upper ends connected to the lower end surface of the fastening nut; and lower ends configured to be in contact with the lower surface of the tub.